Semester Thesis for the Aerospace Engineering Masters at
Technical University of Munich

Analysis of Flamelet Models for High-Pressure Hydrogen Turbulent Combustion

Authors:
Ali Batuhan Akbey, Haydar Ogul Ayas

Supervisors:
Dr. Daniel Martinez Sanchis
Prof. Dr. Agnes Jocher

Institute:
Sustainable Future Mobility

Date:
July 8, 2025

Contents

1 Introduction **3**

2 Methodology **6**
2.1 Average Balance Equations 6
2.2 RANS of Turbulent Non-premixed Flames 8
2.2.1 Implementation of Primitive Variables Method 10
2.3 Problem Definition & Application of Method 17
2.3.1 Problem Definition 17
2.3.2 Application of Method 18

3 Results and Discussion **32**
3.1 Local Evaluation of Predicted PDFs 32
3.2 Local Evaluation of Combined PDFs 34
3.3 Prediction of the Temperature Field 36

4 Conclusion and Future Work **41**

Bibliography **43**

Appendix A **44**

Abstract

This semester thesis focuses on evaluating and enhancing flamelet-based modeling strategies for simulating high-pressure, turbulent hydrogen-oxygen combustion. Such simulations are critical for the design and analysis of propulsion systems, particularly in the context of rocket engines where during the design phase accurate prediction of temperature and species fields is essential.

The study employs the Primitive Variable Method (PVM) within the Reynolds-Averaged Navier-Stokes (RANS) framework to compute Favre-averaged temperature without explicitly solving the full set of equations. Instead, a presumed probability density function (PDF) of the mixture fraction is used to evaluate conditional averages derived from flamelet assumptions.

A central part of the work is the critical examination of the widely used beta-distribution PDF, which is typically constructed using only the first two statistical moments (mean and variance) of the mixture fraction. This classical approach is evaluated by comparing predicted quantities against reference data from a Direct Numerical Simulation (DNS) of a coaxial H_2/O_2 domain. To improve upon the limitations of the standard beta-distribution, particularly its tendency to under and over predict the temperature in some regions of the flow, this study investigates the incorporation of the third statistical moment (skewness) into the beta-distribution with the help of the coefficient c. This coefficient quantifies the contributions of classic beta and mean&skewness distributions.

The methodology is implemented in MATLAB and applied to a 2D domain extracted from the DNS case. The workflow includes pre-processing of DNS-derived statistical moments, construction of PDFs using both classic and skewness beta-functions, calculating an optimal c value for best accuracy and integration of these PDFs with the local flamelet structure to predict Favre-averaged temperatures.

Three key contributions emerge from the work:

- **Enhanced Predictive Accuracy via Higher-Order Moment PDFs:** Inclusion of skewness improves Favre-averaged temperature predictions (particularly near the inlet), highlighting the value of higher-order statistical moments even under limited resolution.
- **Localized Modeling Strategy Based on Flow Structure:** Simulation results indicate that different domain regions favor different PDF constructions: PDF based on mean and variance performs better near pure-fuel zones, while PDF based on mean and skewness offers advantages at flame fronts, suggesting region-specific PDF strategies.
- **Framework for Universal Tabulation of Contribution Coefficients:** The methodology establishes a basis for future tabulation of contribution coefficients (e.g., c_{opt}), potentially via polynomial interpolation, allowing generalized prediction schemes for both temperature and molar fractions in combustion simulations.

The findings underline the sensitivity of flamelet-based methods to PDF assumptions and suggest that incorporating higher-order statistical information can be beneficial in some cases, though further improvements may require coming up with a method to determine a local optimum c coefficient.

1 Introduction

Combustion is one of the most important methods in energy generation since approximately ninety percent of the total energy used in the world is generated through combustion processes [1]. Although there is an increase in the utilization of renewable energy sources, combustion remains the dominant energy generation method. Especially propulsion systems such as rocket engines, aero engines, and gas turbines completely rely on combustion. The efficiency and performance of these systems are directly linked to the characteristics of the combustion process, particularly the flame dynamics, temperature distribution, and heat flux properties.

In propulsion systems, the quality of combustion directly influences the thrust generated, as it determines the energy release rate of the propellants. For instance, for rocket engines, achieving optimal combustion stability, and the control of flame temperature are crucial for maximizing the thrust while minimizing fuel consumption. Given that combustion will play a major role in energy generation and propulsion in the foreseeable future, a comprehensive understanding of its inner workings is crucial.

The combustion phenomenon combines two main disciplines, flow mechanics and chemistry. The complex physics introduced by each discipline and their interaction with each other is the key to understanding combustion. One crucial tool to help researchers achieve this goal is numerical simulation.Although experimental methods have provided—and will continue to provide—essential insights, numerical simulations are an equally indispensable tool for several reasons. Firstly, experiments involving combustion can often be hazardous and pose risks to both personnel and equipment. Simulations, on the other hand, offer a safe alternative by allowing researchers to explore complex combustion characteristics in a virtual environment. Moreover, numerical techniques enable the precise study of conditions which are difficult or impossible to replicate experimentally, such as extreme temperatures, pressures, or transient phenomena. For example, the mixture fraction of specific locations in the combustion chamber is difficult to analyze due to the requirement of extremely complex measurement methods. However, with the accurate simulation of the flame zone, it is possible to achieve mixture fraction map over the entire domain for each time-step of the simulation.

The most accurate and simultaneously the most computationally expensive method for numerical simulation is known as Direct Numerical Simulation (DNS). DNS solves the governing equations numerically without relying on any combustion or turbulence models. This approach captures all scales of the fluid motion and chemical reactions, from the largest eddies to the smallest dissipative scales, such as the Kolmogorov scales in turbulence or the flame thickness in combustion. However, the application of DNS to combustion scenarios comes with significant challenges. Resolving the smallest scales in turbulent reacting flows requires an extremely fine computational grid and very small time steps. This leads to an extremely high computational cost. Therefore, it is compulsory to employ different numerical models, which are time-saving, precise enough, and computationally affordable.

Depending on the specific combustion scenario and computational budget of the user, Reynolds-averaged Navier-Stokes (RANS) or Large Eddy Simulation (LES) can be employed.

LES represents a more advanced computational approach than RANS while having a lower computational cost than DNS. It explicitly resolves the large turbulent scales while modeling the effects of smaller scales using subgrid closure models. This is achieved by filtering the instantaneous balance equations, allowing LES to capture the instantaneous position of a resolved flame front and low-frequency temperature variations [2]. LES has become the standard research tool in the combustion community and is gradually being adopted in industry, particularly for cases where RANS models are known to be imprecise. These include unsteady flows, ignition, quenching, swirling flows, and combustion instabilities.

RANS has historically served as the first practical method for simulating turbulent combustion, since directly resolving the instantaneous flow field within a turbulent flame was computationally infeasible[2]. RANS was developed to solve for the mean values of all quantities by averaging the instantaneous balance equations. These averaged equations, whether using Reynolds or Favre (mass-weighted) averaging, require closure models: a turbulence model to handle flow dynamics and a com-

bustion model to describe chemical species conversion and heat release. For stationary flows, RANS provides time-averaged quantities. For example, in a stabilized flame, RANS predicts a constant mean temperature at a given point, regardless of the temperature history [2]. Despite its limitations, RANS remains the standard approach in commercial combustion codes due to its computational efficiency and robustness for steady-state problems.

Despite the advantages offered by LES and RANS approaches, accurately modeling combustion remains a major challenge due to the high complexity of chemical kinetics and the need to model Turbulence-Chemistry-Interactions(TCI). However, this modeling approach often involves reducing the complex turbulent dynamics to just one or two parameters, which are typically the turbulent kinetic energy and its dissipation rate. This oversimplifies the flow and leads to a loss of important physical details, thereby limiting the ability to accurately capture the underlying processes and impeding reliable TCI modeling [3].

To capture the complex chemistry that is taking place in a turbulent flow, detailed chemical mechanisms, involving a large number of species, are utilized. This necessitates the solution of full set of N-1 chemical species transport equations for N species. A common simplifying assumption for H2/O2 combustion is the chemical equilibrium. This assumption is justified by the fast reaction rate of this propellant combination and the high pressure & temperature conditions it takes place in [4]. As validated and presented by Ivancic et. al [5] and Riedmann et. al [6]this approach delivers accurate results. Although accurate for a large number of cases, capturing non-equilibrium effects and the use of detailed chemical mechanisms necessitate the use of another approach.

In the flamelet concept [7], the turbulent flame, is treated as an ensemble of thin laminar flame structures, so-called flamelets, that are embedded within the turbulent flow field. For flows with large Damköhler numbers, where the flame is thinner than the turbulent scales, this definition is applicable [8]. These flamelets are characterized primarily by the local mixture fraction and its fluctuations. By assuming that the chemical composition and thermodynamic properties can be parameterized by a small set of scalar variables, such as the mixture fraction and its variance, the complexity of the simulation can be drastically reduced while still capturing the essential features of the combustion process.

Within this context, flamelet-based methods like the Primitive Variable Method provide a powerful tool for modeling the detailed chemistry of turbulent flames. Through the Primitive Variable Method, the flame structure is described using conditional quantities. In this approach, the RANS equations are solved only for flow variables (e.g., mean density $\bar{\rho}$, Favre-averaged velocity $\tilde{u}_i$) and mixture fraction variables (e.g., mean mixture fraction $\tilde{z}$ and its higher moments). Once the probability density function $p(z^*)$ is estimated, it is used to determine the species mass fractions $\tilde{Y}_k$ and mean temperature $\tilde{T}$.

However, characterization of the probability density function (PDF) of the mixture fraction in turbulent flows remains an active area of research. In his seminal 1985 work for PDF Methods in turbulent combustion, Pope [9] introduced the transport equations for joint-PDFs of velocity and scalar quantities. By solving the modeled transport equations effects of convection, reaction, and body forces appear exactly and thus do not have to be modeled. This quality of the equations for joint-PDFs makes them a very effective and accurate tool.

In his comprehensive review [10], Haworth discusses the use of beta distributions in presumed PDF models for turbulent combustion, particularly to represent the PDF fraction of the mixture in non-premixed flames. Although the standard beta distribution, determined by mean and variance (standard beta distribution), is widely used for its simplicity and flexibility, it has notable limitations. Specifically, it cannot accurately represent bimodal or complex PDF shapes that emerge in real turbulent flows, especially in systems with spray combustion or intense scalar stratification. Haworth highlights that modified beta distributions, which introduce other variables, can better match these complex shapes as seen in transported PDF (TPDF) results. However, this improvement is offset by the arbitrariness in defining these extra parameters, which undermines the predictive capability.

To address the limitations of the standard and empirically modified beta distributions in modeling turbulent combustion, this thesis explores an alternative approach by incorporating higher-order statistical moments into the scalar PDF representation. By extending the presumed PDF framework beyond just the mean and variance, this work aims to develop a physically grounded, generalizable model that better captures the non-Gaussian, multimodal distributions often encountered in high-pressure hydrogen combustion. Motivated by the shortcomings identified in Haworth's review —specifically the inadequacy of beta distributions in capturing the true shape of scalar fields— this study seeks to improve the accuracy of flamelet models and enhance their applicability to realistic engine-relevant conditions, where TCIs are both strong and highly nonlinear.

To validate the accuracy of the predicted species mass fractions $\tilde{Y}_k$ and mean temperature $\tilde{T}$, the results will be compared with precomputed Direct Numerical Simulation (DNS) results of the same flame structure. The primary aim of this paper is to assess the accuracy of the prediction of these variables, in terms of the used β-PDF function. Furthermore, modifications to the Primitive Variable Method will be discussed to enhance the accuracy of the predictions.

2 Methodology

2.1 Average Balance Equations

To enable Reynolds-Averaged Navier–Stokes (RANS) simulations, the instantaneous governing equations must be averaged to express them in terms of mean quantities. However, this averaging process introduces additional terms—known as closure problems—that cannot be directly computed and therefore require modeling, typically through the use of turbulence and combustion models.

The governing balance equations for mass, species, momentum, and enthalpy, which serve as the foundation for the simulation, can be written respectively as:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i) = 0 \tag{1}$$

$$\frac{\partial \rho u_j}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i u_j) + \frac{\partial p}{\partial x_j} = \frac{\partial \tau_{ij}}{\partial x_i} \tag{2}$$

$$\frac{\partial (\rho Y_k)}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i Y_k) = -\frac{\partial}{\partial x_i}(V_{k,i} Y_k) + \dot{\omega}_k \quad \text{for} \quad k = 1, N \tag{3}$$

$$\frac{\partial \rho h_s}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i h_s) = \dot{\omega}_T + \frac{Dp}{Dt} + \frac{\partial}{\partial x_i}\left(\lambda \frac{\partial T}{\partial x_i}\right) - \frac{\partial}{\partial x_i}\left(\rho \sum_{k=1}^{N} V_{k,i} Y_k h_{s,k}\right) + \tau_{ij}\frac{\partial u_i}{\partial x_j} \tag{4}$$

For an incompressible flow, Reynolds averaging splits each quantity f into a mean $\overline{f}$ and fluctuating f' component ($f = \overline{f} + f'$). Using Reynolds averaging, the mass conservation Equation 1 transforms into the following equation.

$$\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial}{\partial x_i}(\overline{\rho u_i}) = \frac{\partial \bar{\rho}}{\partial t} + \frac{\partial}{\partial x_i}\left(\overline{\rho}\,\overline{u_i} + \overline{\rho' u_i'}\right) = 0 \quad \text{or} \quad \frac{\partial \bar{\rho}}{\partial t} + \frac{\partial}{\partial x_i}\left(\overline{\rho}\,\overline{u_i}\right) = -\frac{\partial}{\partial x_i}\left(\overline{\rho' u_i'}\right) \tag{5}$$

However, the unclosed quantity $\overline{\rho' u_i'}$ corresponding to the correlation between density and velocity fluctuations appears and this term has to be modeled, which presents an additional challenge. Moreover, this correlation term also acts as a mass source term for the mean flow field, which can be problematic for CFD simulations. For instance, in steady-state flows, the average mass flow rates calculated using Reynolds averaging might not be conserved, which is counterintuitive and difficult to manage.

To address these issues, especially in flows with variable density, the mass weighted averages (known as Favre averages) are often used. Favre averaging helps to avoid complications introduced by the correlation terms and simplifies the modeling process. This approach is preferred in many practical applications. The Favre average is defined as:

$$\widetilde{f} = \frac{\overline{\rho f}}{\overline{\rho}} \tag{6}$$

In analogy to the Reynolds average, any quantity f may be split into mean and fluctuating components as:

$$f = \widetilde{f} + f'' \quad with \quad \widetilde{f''} = 0 \tag{7}$$

By using aforementioned formalism, the averaged balance equations become:

Mass:

$$\frac{\partial \overline{\rho}}{\partial t} + \frac{\partial}{\partial x_i}(\overline{\rho}\widetilde{u_i}) = 0 \tag{8}$$

Momentum:

$$\frac{\partial \overline{\rho}\widetilde{u_i}}{\partial t} + \frac{\partial}{\partial x_i}(\overline{\rho}\widetilde{u_i}\widetilde{u_j}) + \frac{\partial \overline{p}}{\partial x_j} = \frac{\partial}{\partial x_i}(\overline{\tau}_{ij} - \overline{\rho}\widetilde{u_i''u_j''}) \tag{9}$$

Chemical Species:

$$\frac{\partial(\overline{\rho}\widetilde{Y_k})}{\partial t} + \frac{\partial}{\partial x_i}(\overline{\rho}\widetilde{u_i}\widetilde{Y_k}) = -\frac{\partial}{\partial x_i}(\overline{V_{k,i}Y_k} + \overline{\rho}\widetilde{u_i''Y_k''}) + \dot{\omega}_k \quad \text{for} \quad k = 1, N \tag{10}$$

Enthalpy:

$$\frac{\partial \overline{\rho}\widetilde{h_s}}{\partial t} + \frac{\partial}{\partial x_i}(\overline{\rho}\widetilde{u_i}\widetilde{h_s}) = \dot{\omega_T} + \overline{\frac{Dp}{Dt}} + \frac{\partial}{\partial x_i}\left(\overline{\lambda\frac{\partial T}{\partial x_i}} - \overline{\rho u_i'' h_s''}\right) + \overline{\tau_{ij}\frac{\partial u_i}{\partial x_j}} - \frac{\partial}{\partial x_i}\overline{(\rho\sum_{k=1}^{N} V_{k,i}Y_k h_{s,k})} \tag{11}$$

where

$$\overline{\frac{Dp}{Dt}} = \frac{\partial \overline{p}}{\partial t} + \overline{u_i\frac{\partial p}{\partial x_i}} = \frac{\partial \overline{p}}{\partial t} + \widetilde{u_i}\frac{\partial \overline{p}}{\partial x_i} + \overline{u_i''\frac{\partial p}{\partial x_i}}$$

These equations are formally identical to the classical Reynolds averaged equation where the density of the flow is assumed to be constant.

The Favre-averaged balance equations contain several unclosed terms, and the primary goal of turbulent combustion modeling is to develop closures for these unknown quantities, as outlined in Equation 8 to Equation 11. Depending on the specific turbulent combustion scenario, the following terms require modeling: Reynolds stresses ($\widetilde{u_i''u_j''}$), turbulent fluxes of species ($\widetilde{u_i''Y_k''}$) and enthalpy ($\widetilde{u_i''h''}$), laminar diffusive fluxes, chemical reaction rates of species ($\overline{\dot{\omega}_k}$), and pressure-velocity correlations ($\overline{u_i''\frac{\partial p}{\partial x_i}}$). While Reynolds stresses are typically addressed using classical turbulence models developed for non-reacting flows, the remaining terms necessitate case-specific approaches to achieve closure.

2.2 RANS of Turbulent Non-premixed Flames

For the case of turbulent non-premixed flames, most of the calculations and assumptions rely on the mixture fraction z. Pre-introduced averaged balance equations get even simpler forms with the introduction of mixture fraction. According to the [2], the following assumptions are used for the derivation of classical models for the turbulent non-premixed flames:

1. The thermodynamic pressure is constant and Mach numbers are small.
2. Species heat capacities are equal and constant ($C_{ph} = C_p$).
3. Molecular diffusion follows Fick's law and molecular diffusivities D_k are equal for all species ($D_k = \mathcal{D}$).
4. Lewis numbers are equal to unity ($Le = \frac{\lambda}{\rho C_p \mathcal{D}} = 1$).
5. Fuel and oxidizer streams are separately introduced into the combustion chamber with reference state (T_F^0, Y_F^0) for fuel and (T_O^0, Y_O^0) for oxidizer.

According to the last assumption, for a single one-step chemistry in adiabatic flows, the mixture fraction variable z can be introduced as:

$$z = \frac{sY_F - Y_O + Y_O^0}{sY_F^0 + Y_O^0} \tag{12}$$

The value s in the Equation Equation 12 is the stoichiometric oxidizer to fuel ratio, depending on the selected propellant.

Following the introduction of the mixture fraction variable z, the system of balance equations is simplified to:

$$\frac{\partial \overline{\rho}}{\partial t} + \frac{\partial}{\partial x_i}(\overline{\rho}\widetilde{u}_i) = 0 \tag{13}$$

$$\frac{\partial \overline{\rho}\widetilde{u}_i}{\partial t} + \frac{\partial}{\partial x_j}(\overline{\rho}\widetilde{u}_i\widetilde{u}_j) + \frac{\partial \overline{p}}{\partial x_j} = \frac{\partial}{\partial x_i}\left(\overline{\tau}_{ij} - \rho\widetilde{u_i''u_j''}\right) \tag{14}$$

$$\frac{\partial(\overline{\rho}\widetilde{Y}_k)}{\partial t} + \frac{\partial}{\partial x_i}(\overline{\rho}\widetilde{u}_i\widetilde{Y}_k) = \frac{\partial}{\partial x_i}\left(\overline{\rho\mathcal{D}_k\frac{\partial Y_k}{\partial x_i}} - \overline{\rho}\widetilde{u_i''Y_k''}\right) + \overline{\dot{\omega}_k} \quad \text{for} \quad k = 1, N \tag{15}$$

$$\frac{\partial \overline{\rho}\widetilde{z}}{\partial t} + \frac{\partial}{\partial x_i}(\overline{\rho}\widetilde{u}_i\widetilde{z}) = \frac{\partial}{\partial x_i}\left(\overline{\rho\mathcal{D}\frac{\partial z}{\partial x_i}} - \overline{\rho}\widetilde{u_i''z''}\right) \tag{16}$$

As mentioned before, this set of equations contains some unclosed terms: Reynolds Stresses ($\widetilde{u_i''u_j''}$) are closed using classical turbulence models for non-reacting flows, and turbulent scalar transport ($\widetilde{u_i''Y_k''}$ and $\widetilde{u_i''z''}$) is usually modeled by the gradient assumption.

For adiabatic flows with single-step chemistry, fuel and oxidizer mass fractions, temperature and mixture fraction are related through Equation Equation 12, therefore, the typical method is solving the

set of equations in terms of mean fuel mass fraction $\widetilde{Y_F}$ and mixture fraction $\widetilde{z}$ with the appropriate models for unclosed terms. Mean oxidizer mass fraction and temperature $\widetilde{T}$ are then calculated from $\widetilde{Y_F}$ and $\widetilde{z}$ by using Equation Equation 12.

Turbulence adds an additional complexity due to the averaging procedures. In order to determine average values, the mean values of mixture fraction variable is not sufficient. The full Probabilty Density Function (PDF) of z is required. As stated in [2], When the PDF of z, $p(z)$ is known, the average species mass fraction $\widetilde{Y_k}$, average temperature $\widetilde{T}$, and average reaction rate $\overline{\dot{\omega}_k}$ can be calculated as:

$$\bar{\rho}\widetilde{Y_k} = \int_0^1 \left(\overline{\rho Y_k | z^*}\right) p(z^*)\, dz^* \tag{17}$$

$$\bar{\rho}\widetilde{T} = \int_0^1 \left(\overline{\rho T | z^*}\right) p(z^*)\, dz^* \tag{18}$$

$$\overline{\dot{\omega}}_k = \int_0^1 \left(\overline{\dot{\omega}_k | z^*}\right) p(z^*)\, dz^* \tag{19}$$

In the equations listed above, $\overline{(Q|z^*)}$ represents the conditional average quantity of Q for a given value of mixture fraction $z = z^*$, and $p(z^*)$ denotes the z-probability density function (PDF).

In the literature, two primary approaches are commonly used to model turbulent non-premixed flames.

Primitive Variable Method

This method is based on Equation Equation 17 and Equation 18. The aim of Primitive Variable Method is to provide the conditional quantities $\overline{(\rho Y_k|z^*)}$ and $\overline{(\rho T|z^*)}$. Once these conditional quantities are obtained, the average values for temperature $\widetilde{T}$ and species mass fraction $\widetilde{Y_k}$ can be calculated. As a result, species mass fraction equation (Equation Equation 15) and temperature balance equations are no longer required and mean reaction rates $\overline{\dot{\omega}_k}$ do not need to be modeled explicitly. In RANS codes, the equations are solved only for flow variables and mixture fraction variables (along with their higher moments) to make an estimation for probability density function $p(z^*)$ throughout the domain. This method will be described and applied in detail throughout this paper.

Reaction Rate Approach

This method can be considered the exact opposite of the Primitive Variable Method. The balance equations for species mass fraction (Equation Equation 15) and for temperature are solved. Unlike the primitive variable method, reaction rates $\overline{\dot{\omega}_k}$ are modeled for each reaction.

The primitive variable method is significantly less time-consuming compared to the reaction rate approach, since it eliminates species mass fraction and temperature balance equations. Conversely, reaction rate approach is more accurate as it includes the additional factors such as compressibility, heat losses, or secondary reactant injections. In the primitive variable method, mean species mass fractions $\widetilde{Y_k}$ and mean temperature $\widetilde{T}$ are directly calculated from $\overline{(\rho Y_k|z^*)}$ and $\overline{(\rho T|z^*)}$ by using Equations Equation 17 and Equation 18. In this approach, since the balance equation for species mass fraction

is eliminated, there is no need to model the turbulent fluxes anymore.

Given that this paper focuses on the application of the primitive variable method, a more detailed discussion of its implementation and implications will be provided.

2.2.1 Implementation of Primitive Variables Method

For infinitely fast irreversible chemistry with unity Lewis number and adiabatic combustion case, instantaneous values of species mass fractions Y_k and temperature T are directly linked to the mixture fraction z through the relations $Y_k(z)$ and $T(z)$.

The flame structure can be defined either through precomputed flamelet libraries or derived from experimental or simulation data. Flamelet libraries available in the literature typically represent the flame in terms of the mixture fraction, using empirical data. Alternatively, for case-specific scenarios, a custom flamelet library can be generated based on high-resolution experimental measurements. Finally, instead of relying on experimental data, Direct Numerical Simulation (DNS) can be employed to construct the flamelet library; however, the high computational cost and the required level of accuracy must be carefully considered when selecting the most suitable method.

For infinitely fast chemistry, the conditional averages directly calculated from $\overline{(\rho Y_k|z^*)}$ and $\overline{(\rho T|z^*)}$ in Equations Equation 17 and Equation 18 reduce to:

$$\left(\overline{\rho Y_k|z^*}\right) = \rho\left(z^*\right) Y_k\left(z^*\right) \tag{20}$$

$$\left(\overline{\rho T|z^*}\right) = \rho\left(z^*\right) T\left(z^*\right) \tag{21}$$

By using these conditional averages, mean species mass fractions and temperature values are obtained from:

$$\overline{\rho}\widetilde{Y_k} = \int_0^1 \rho(z^*) Y_k(z^*) p(z^*)\, dz^* \tag{22}$$

$$\overline{\rho}\widetilde{T} = \int_0^1 \rho(z^*) T(z^*) p(z^*)\, dz^* \tag{23}$$

Additionally, the probability density functions can be expressed as mass-weighted probability density functions as: $\widetilde{p}(z^*) = \rho(z^*)p(z^*)/\overline{\rho}$, as a result mean species mass fraction and temperature values become:

$$\widetilde{Y_k} = \int_0^1 Y_k(z^*)\widetilde{p}(z^*)\, dz^* \tag{24}$$

$$\widetilde{T} = \int_0^1 T(z^*)\widetilde{p}(z^*)\, dz^* \tag{25}$$

Consequently, the evaluation of $\widetilde{Y_k}$ and $\widetilde{T}$ depends on accurately determining the probability density function $\widetilde{p}(z)$ of the mixture fraction z. Estimating this PDF remains one of the central challenges in the modeling of turbulent flows. This paper will address this challenge by proposing potential solutions for modeling the PDF. Various methods will be examined, and their accuracy will be assessed to identify the most effective approach.

Probability Density Function (PDF)

A probability density function (PDF) describes the likelihood of a continuous random variable taking on a specific value within its sample space. It provides a relative likelihood of the variable being close to a particular value, rather than the absolute probability (which is zero for continuous variables).In the context of the primitive variable method, the probability density function (PDF) of the mixture fraction variable z represents the likelihood of z taking a specific value z^* at a given point in the domain. For each cell in the domain, z must lie within the interval $0 \leq z \leq 1$, where $z = 0$ corresponds to pure oxidizer and $z = 1$ corresponds to pure fuel. The PDF of z quantifies the likelihood of each possible z value at that specific location.

The flame structure can be defined using either flamelet libraries or experimental (or simulation) results. Flamelet libraries, available in the literature, describe the flame in terms of the mixture fraction based on empirical data. For case-specific scenarios, similar flamelet libraries can be obtained through experiments using high-resolution measurement techniques. Alternatively, Direct Numerical Simulation (DNS) can be employed to generate flamelet libraries without relying on experimental data. However, when choosing the optimal method, the computational cost of DNS and the required accuracy of the flamelet library must be carefully considered.

The β function is the most popular presumed PDF function for the turbulent flames and depends only on two parameters. For the most general applications, β function depends on the mean mixture fraction $\widetilde{z}$ and its variance $\widetilde{z''^2}$.

$$p(\Theta) = \frac{1}{B(a,b)}\Theta^{a-1}(1-\Theta)^{b-1} = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\Theta^{a-1}(1-\Theta)^{b-1} \tag{26}$$

where $B(a,b)$ is the normalization factor and it can be expressed in terms of the Γ function.

The normalization factor $B(a,b)$ and the Γ function can be defined as:

$$B(a,b) = \int_0^1 \Theta^{a-1}(1-\Theta)^{b-1}\, d\Theta \tag{27}$$

$$\Gamma(x) = \int_0^{+\infty} e^{-t}t^{x-1}\, dt \tag{28}$$

In the Gamma function, t is the integration variable. It is a dummy variable to use perform the integration over the range $[0, +\infty)$.

In order to obtain, β-function the parameters of a and b has to be determined by plugging Eq 26 into definitions of first two moments of the progress variable $\widetilde{\Theta}$:

$$\widetilde{\Theta} = \int_0^1 \Theta^* \widetilde{p}(\Theta^*)\, d\Theta^* \tag{29}$$

$$\widetilde{\Theta''^2} = \int_0^1 \left(\Theta^* - \widetilde{\Theta}\right)^2 \widetilde{p}(\Theta^*)\, d\Theta^* \tag{30}$$

leads to:

$$a = \widetilde{\Theta}\left[\frac{\widetilde{\Theta}(1-\widetilde{\Theta})}{\widetilde{\Theta''^2}}\right] \tag{31}$$

$$b = \frac{a}{\widetilde{\Theta}} - a \tag{32}$$

These equations describe the construction of the β-function (or β-PDF) in terms of a specific progress variable Θ. For turbulent combustion problems, the relevant progress variable is the mixture fraction variable z. Therefore, Equations Equation 26 to Equation 32 can be expressed in terms of the mixture fraction variable z as follows:

$$p(z) = \frac{1}{B(a,b)} z^{a-1}(1-z)^{b-1} = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} z^{a-1}(1-z)^{b-1} \tag{33}$$

where

$$B(a,b) = \int_0^1 z^{a-1}(1-z)^{b-1}\, dz \tag{34}$$

$$\Gamma(x) = \int_0^{+\infty} e^{-t} t^{x-1}\, dt \tag{35}$$

Additionally, the first two moment of the mixture fraction variable z can be expressed as:

$$\widetilde{z} = \int_0^1 z^* \widetilde{p}(z^*)\, dz^* \tag{36}$$

$$\widetilde{z''^2} = \int_0^1 (z^* - \widetilde{z})^2 \widetilde{p}(z^*)\, dz^* \tag{37}$$

which also leads to:

$$a = \widetilde{z}\left[\frac{\widetilde{z}(1-\widetilde{z})}{\widetilde{z''^2}} - 1\right] \tag{38}$$

$$b = \frac{a}{\widetilde{z}} - a \tag{39}$$

Once the β-PDF of the mixture fraction variable z is determined in terms of its average value $\widetilde{z}$ and its higher moment (variance $\widetilde{z''^2}$) using the aforementioned formulas, the mean temperature $\widetilde{T}$ and mean species mass fractions $\widetilde{Y_k}$ can be calculated by using Equations Equation 24 and Equation 25. To further understand the process, an example case can be considered to illustrate how to calculate the mean temperature value $\widetilde{T}$. Since the flame structure is already defined in terms of $T(z)$ and $Y_k(z)$, and the function $T(z)$ is not continuous, Equations Equation 24 and Equation 25 can be implemented in a non-continuous form. Instead of using an integral over the range $[0, 1]$, the calculation is performed by summing the contributions of $T(z)$ at discrete z values. This approach involves evaluating $T(z)$ at a finite number of z points and summing their weighted contributions, as follows:

z	0	0.2	0.4	0.6	0.8	1.0
$T(z)$	300	3400	3000	1500	1000	300
$p(z)$	0.1	0.2	0.3	0.2	0.1	0.1

Table 1: Table showing hypothetical discrete values of z, $T(z)$, and $p(z)$.

Table 1 represents the hypothetical probability values of the mixture fraction variable z at a specific point in the domain. The temperature $T(z)$ is obtained from either flamelet libraries or experimental results. For example, in the second column, the probability of the mixture fraction z being 0.2 at this point is 20%. Additionally, this implies that if $z = 0.2$, the temperature at this point must be 3400 K.

By using this discretization, $\widetilde{T}$ can be calculated as:

$$\widetilde{T} = \frac{\sum T(z).p(z)}{\sum p(z)} \tag{40}$$

$$\widetilde{T} = \frac{T_1.p(z_1) + T_2.p(z_2) + T_3.p(z_3) + T_4.p(z_4) + T_5.p(z_5) + T_6.p(z_6)}{p(z_1) + p(z_2) + p(z_3) + p(z_4) + p(z_5) + p(z_6)} \tag{41}$$

By plugging the hypothetical values in the Table 1 into the Equations Equation 40 or Equation 41:

$$\widetilde{T} = \frac{(300).(0.1) + (3400).(0.2) + (3000).(0.3) + (1500).(0.2) + (1000).(0.1) + (300).(0.1)}{0.1 + 0.2 + 0.3 + 0.2 + 0.1 + 0.1} = 2400K \tag{42}$$

In the Equation Equation 42, the calculation of the average temperature of the cell throughout the simulation is illustrated with the hypothetical z, $p(z)$ and $T(z)$ values.

This method is known as the **Primitive Variables Method** as the calculation of all mean variables rely on the the direct calculation of the Favre average of the temperature $\widetilde{T}$. In a transient simulation, the mean mixture fraction $\widetilde{z}$ and its variance $\widetilde{z''^2}$ are obtained using a RANS code. Each cell in the computational domain has its own $\widetilde{z}$ and $\widetilde{z''^2}$ values. Using these values, the β-PDF for a specific cell is calculated using Equations Equation 33 to Equation 39. In a transient simulation, the mixture fraction z typically varies over time. However, each z value contributes to the calculation of $\widetilde{T}$ with a corresponding weight, represented by the probability density function $p(z)$. The same procedure can also be applied to compute the average species mass fraction $\widetilde{Y_k}$. By employing this method, there is no need to solve the species mass fraction equation explicitly anymore, which significantly reduces the computational cost and saves simulation time.

Limitations & Key Points to be Addressed

First, if the mixture fraction variable z attains the value $z = 0$ or $z = 1$, the normalization factor B becomes zero. This results in an undefined expression because, in Equation Equation 33, the denominator of the expression becomes zero. In order to avoid this undefined expression, the interval for the mixture fraction variable z should be defined as $0 < z < 1$ instead of $0 \leq z \leq 1$.

Secondly the variance of the average mixture fraction variable might attain the value $\widetilde{z''^2} = 0$. This also results in an undefined expression since in Equation Equation 38, the denominator of the coefficient a becomes zero. Statistically, if $\widetilde{z''^2} = 0$, the variance of the mixture fraction variable is zero which means the mixture fraction variable z has to be constant throughtout the simulation. Moreover, theoretically, mixture fraction variable can be only constant for the infinitesimmaly close vicinity of the injection element where there is no expectation of time-dependent mixing . Since the case is non-premixed turbulent combustion, in the infinitesimally close vicinity of the injection element, the fluid can be either pure fuel or pure oxidizer. Therefore, if $\widetilde{z''^2} = 0$, $z = \widetilde{z}$ it must be either $z = \widetilde{z} = 0$ as pure oxidizer or $z = \widetilde{z} = 1$ as pure fuel.

Finally, the overall implementation of the primitive variable method by including all relevant sections is displayed in Figure Figure 1.

Mean Variables Used in RANS Code

Dynamic Pressure, $\widetilde{u}$, $\widetilde{v}$, $\widetilde{w}$, $\widetilde{z}$, $\widetilde{z''^2}$ values are calculated from the balance equations (Equations Equation 13, Equation 14, Equation 16) by using RANS code. However, the solution of these equations requires additional inputs such as Turbulence Variables, and $\overline{\rho}$. Assuming that, $k - \epsilon$ model is used, turbulence variables ,which are used the describe Reynolds stresses and turbulent fluxes, are the turbulent kinetic energy k and the turbulence dissipation rate ϵ.

On the other hand, average mixture fraction $\widetilde{z}$ and its variance $\widetilde{z''^2}$ are calculated using Equation Equation 16. One should note that, in order to calculate the $\widetilde{z''^2}$, additional continuity equation for $\widetilde{z''^2}$ shall be derived as well. To solve this equation for required variables, the balance equations must be closed with appropriate modeling. However, this aspect is out of scope of this paper since $\widetilde{z}$ and $\widetilde{z''^2}$ values are directly retrieved from DNS results to reduce the work load. Since the mixture fraction value z is computed for each cell on the domain by DNS, $\widetilde{z}$ and $\widetilde{z''^2}$ are statistically derived to be utilized in Primitive Variable Method.

One should note that, in order to solve the balance equations listed as Equations Equation 13, Equa-

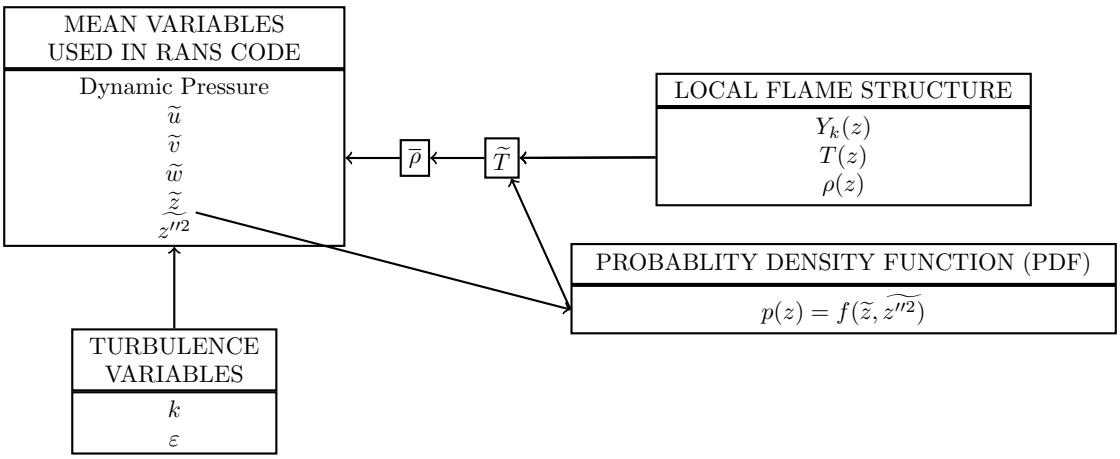


Figure 1: Schematic representation of the solution methodology for turbulent premixed combustion using primitive variable methods, incorporating mean variables, turbulence modeling, and presumed probability density functions.

tion 14, and Equation 16, the mean value of density (or reynolds average of density) $\overline{\rho}$ has to be used. This value is obtained from the flame structure and the mixture fraction probability density function.

Local Flame Structure

Local flame structure, often called as flamelet libraries, is used to define the flame variables such as species mass fractions Y_k, temperature T, and density ρ as functions of the mixture fraction variable z. As previously mentioned, these relationships can be attained from experimental data, simulation results, or empirical correlations. In the context of this paper, the local flame structure is constructed based on prior experimental results. Specifically, $T(z)$ and $Y_k(z)$ (only for single product) are provided and utilized in calculations.

The Fig. Figure 2 and Figure 3 visualize the relationship between the single look-up variable, mixture fraction, with adiabatic flame temperature and molar fraction of the product specie H_2O. The temperature values at 0 and 1 represent the injection temperatures of oxidizer and fuel respectively. The adiabatic flame temperature peaks at around the stoichiometric mixture fraction 0,1 while the highest H_2O production occurs at a richer mixture. Molar fraction curve behaves quite linearly in both rich and lean regions. Adiabatic flame temperature on the other hand, reduces in a non-linear fashion in the rich region. It is also important to note that the oxidizer-rich region until $Z = 0,1$ has a very high temperature gradient and this needs to be taken into account during discretization of the Z-T domain. A low resolution in this region can lead to disproportionally high errors in the temperature prediction.

While the mixture fraction z is commonly used as the sole look-up variable in simplified flamelet models, the accuracy of the local flame structure can be significantly improved by including additional parameters. In more complex flow environments, ex. trans-critical injection, strongly cooled walls, relying only on z cannot capture all relevant physical effects[11].

To account for these effects, advanced flamelet libraries can include additional look-up variables such as pressure p, scalar dissipation rate χ, or enthalpy h. These extensions allow for a more accurate representation of the flame structure under varying conditions:

- **Pressure**: Including pressure as a look-up variable is essential in high-pressure combustion systems, especially in scenarios involving trans-critical injection or strong pressure gradients, such as during expansion in the nozzle.

- **Scalar Dissipation Rate (χ)**: The scalar dissipation rate quantifies the rate of mixing between fuel and oxidizer. In non-premixed combustion, it plays a critical role in extinction and re-ignition, thus unsteady flame phenomena. Incorporating χ enables better modeling of these localized flame dynamics.

- **Enthalpy or Temperature**: In cases with strong heat losses, such as near cooled walls or regenerative cooling channels, the local flame temperature can deviate significantly from the adiabatic assumption. Including enthalpy as a variable is a way of including non-adiabatic effects in the calculations.

By expanding the dimensionality of the flamelet library, these advanced multi-parameter models can more accurately reflect the local thermochemical state, enabling improved prediction of combustion behavior in critical regions such as the injector, chamber walls, and nozzle throat. However, this increase in complexity comes at the cost of greater memory requirements and interpolation effort, which must be managed in numerical implementations.

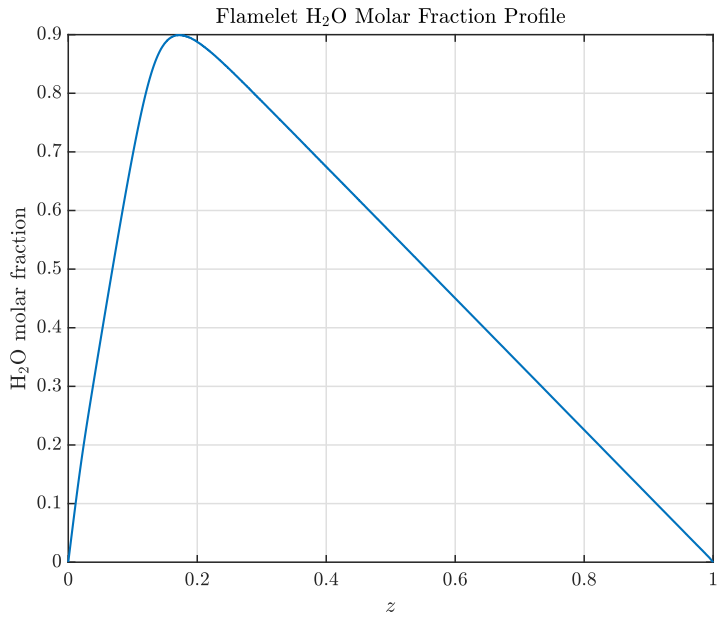


Figure 2: Flamelet H_2O Molar Fraction Profile

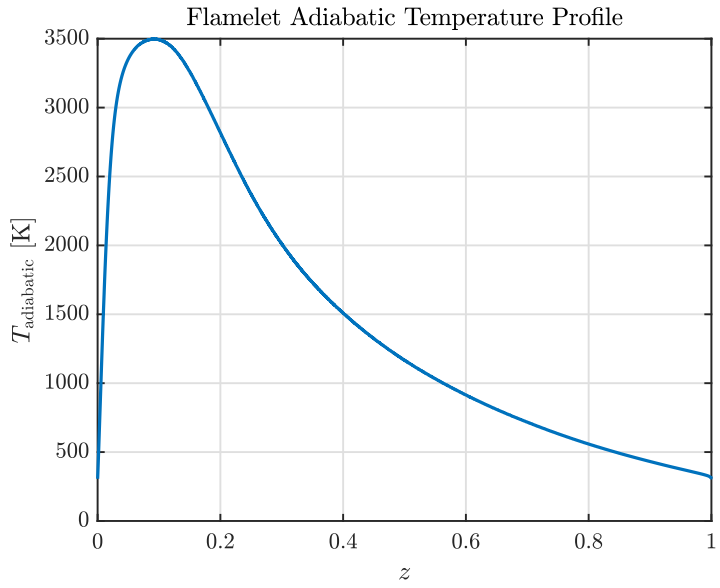


Figure 3: Flamelet Adiabatic Flame Temperature Profile

Probablity Density Function (PDF)

Probability density function (PDF) is calculated by using mean value of mixture fraction $\widetilde{z}$ and its variance $\widetilde{z''^2}$. The methodology for constructing the PDF has been described earlier in the paper.

2.3 Problem Definition & Application of Method

2.3.1 Problem Definition

This paper focuses on the practical implementation of a theoretical method for modeling non-premixed turbulent combustion, validated using existing direct numerical simulations results for a specific case. The primary objective is to demonstrate the application of the primitive variable method to calculate relevant flow and combustion variables, with particular emphasis on the Favre-averaged temperature $\widetilde{T}$ and species mass fraction $\widetilde{Y_k}$.

The DNS study focuses on non-premixed turbulent combustion in a rocket engine, where propellants are Hydrogen H_2 and Oxygen O_2, injected coaxially through a single injector element. To minimize computational cost and effort, the simulation is conducted in two dimensions (2D), considering only one injector element. Additionally, the domain is assumed to be symmetric about its central axis, further simplifying the computational model. Although the simulation is transient and spans a long time interval, the data provided for this study consists solely of Favre-averaged values of the relevant variables.

The computational domain for the simulation comprises a grid of 1096×6232 structured quadrilateral cells. In the vertical (x-axis) direction, there are 1096 cells, with the first 548 cells exhibiting symmetry about the central axis. In the horizontal z-axis direction, domain consists of 6232 cells, with no symmetry.

In this paper, a specific aspect of the primitive variable method will be explained in detail. The overall solution methodology for the calculation of all mean variables of the flow is illustrated in the Figure 1. As mentioned previously, the primary scope of this work is the calculation of the Favre-averaged temperature $\widetilde{T}$ and the Favre-averaged species mass fraction $\widetilde{Y_k}$. The accuracy of the predicted Favre-averaged temperature $\widetilde{T}$ is inherently depends to the probability density function used to describe the flow. Therefore, the precision of $\widetilde{T}$ depends on how well the PDF captures the statistical distribution of the mixture fraction z. Once the accuracy of the PDF sufficiently accurate, further improvements in the accuracy of $\widetilde{T}$ can only be achieved by increasing the resolution of the mixture fraction domain

z during the numerical integration process.

2.3.2 Application of Method

The methodology for calculating the Favre-averaged temperature $\widetilde{T}$ is provided in the Equation Equation 25. To compute the $\widetilde{T}$, it is first necessary to construct a probability density function for the mixture fraction variable z, denoted as $p(z$, which depends on $\widetilde{z}$ and $\widetilde{z''^2}$. For turbulent non-premixed combustion cases, where primitive variable method is applied under the assumption of infinitely fast chemistry, the $\widetilde{z}$ and $\widetilde{z''^2}$ are typically determined by solving Equation Equation 16 and an additional continuity equation for $\widetilde{z''^2}$. However, in this study, to focus on the calculation of $\widetilde{T}$ and $\widetilde{Y_k}$, the determination of the values $\widetilde{z}$ and $\widetilde{z''^2}$ are bypassed and these values are directly obtained from DNS results. During DNS, the value of $\widetilde{z}$ and its higher-order moments, such as variance, skewness, and kurtosis are already computed statistically and provided.

Normally, β-function is the most widely used presumed PDF function for turbulent flames, particularly in the most general applications. The β function depends on two parameters: specifically and most generally mean mixture fraction $\widetilde{z}$ and its variance $\widetilde{z''^2}$. However, in this study, higher-order moments such as skewness (3rd order) and kurtosis (4th order) also provided, as their influence on the accuracy of the PDF construction will be investigated. By incorporating these higher-order moments, the study aims to evaluate their impact on the precision of the PDF and consequently, on the accuracy of the computed Favre-averaged temperature $\widetilde{T}$ and species mass fraction $\widetilde{Y_k}$ variables.

Initial Step: Visualization of the Mixture Fraction PDF and Prediction of Favre-Averaged Temperature for a Cell in the Domain

The necessary calculations are implemented using MATLAB. As a first step, a script has been developed to visualize the probability density function of the mixture fraction variable z at a specific coordinate. This initial script serves as a foundational tool for analyzing the distribution of the mixture fraction and provides insights into the statistical behavior of the flow.

The β-function is constructed in MATLAB using the built-in function $betapdf(x, a, b)$, which implements Eq. Equation 33 with the same parameters a and b as defined in Equations Equation 33 and Equation 34. Here, x represents the mixture fraction value for which the probability density function is evaluated. In the initial script, the PDF of the mixture fraction variable z is calculated at a specific coordinate. As previously mentioned, the mean mixture fraction $\widetilde{z}$ and its variance $\widetilde{z''^2}$ are provided directly from the DNS results. To execute the *betapdf* function, the parameters a and b must first be calculated using Equations Equation 38 and Equation 39.

The only discrepancy encountered during the calculation of *betapdf* arises for certain values of $\widetilde{z''^2}$ which makes the constant a being undefined (NaN). To address this issue, a small adjustment is made on the script.

First, it is important to note that if the value of the constant a is computed as, not a number (NaN), this indicates that the variance $\widetilde{z''^2}$ at that location has to be zero. Variance is a measure of dispersion, which determines how for a set of numbers is deviates from its average value. Therefore, if the variance of a distribution is zero, it implies that all realizations of the random variable are constant and equal to the mean value. Therefore, for these specific points where $\widetilde{z''^2} = 0$, the mean mixture fraction $\widetilde{z}$ must be constant and can only take values of 0 (pure oxidizer) or 1 (pure fuel).

The above argument can be both validated theoretically and verified through DNS data. Theoretically, if the variance value $\widetilde{z''^2}$ is zero for time-averaged results, it implies that there is no mixing between fuel and oxidizer. However, in non-premixed turbulent combustion, the continuous motion of the eddies ensures that mixing throughout simulation. Therefore, the only location where mixing

can be theoretically zero can theoretically only possible is at the exact tip of the injection element. As flow progresses downstream, fuel and oxidizer mix more extensively due to the increasing size of eddies which driven by turbulence. Consequently, the dispersion of values from their mean increases in the flow direction, reflecting enhanced mixing due to the growing size of turbulent eddies.

In the provided $\widetilde{z''^2}$ matrix, there are no points where $\widetilde{z''^2}$ is exactly zero. This is because capturing the exact tip of the injection element would require the theoretical dimensions of the cells to be precisely zero, which is not feasible in practice. However, there are numerous points in the domain where $\widetilde{z''^2}$ attains values that are significantly close to zero. These points are also problematic because the precision of the software is insufficient to compute the parameter a when the denominator of Equation Equation 33 approaches very low values (close to zero). Without the aforementioned adjustment, the value of a for these points would be computed as either undefined (NaN) or infinitely large, leading to an undefined (NaN) or incorrect β-PDF distribution.

The relevant threshold for $\widetilde{z''^2}$ is selected as 10^{-8}, with the condition defined as $\widetilde{z''^2} < 10^{-8}$. In the script, for all points in the domain where $\widetilde{z''^2} < 10^{-8}$, the variance is adjusted to $\widetilde{z''^2} = 0$. The script identifies these locations and assigns the mean mixture fraction $\widetilde{z}$ a value of either 0 or 1 with 100% probability. Since $p(z) = 1$, the only two possibilities are $\widetilde{z} = z = 1$ or $\widetilde{z} = z = 0$. Consequently, the β-PDF distribution at these points takes the form of a Dirac-delta-like function, sharply peaked at either $z = 1$ or $z = 0$ for the given z interval from 0 to 1.

In Figure Figure 5, the cells where the variance $\widetilde{z''^2}$ is less than 10^{-8} are highlighted in red. The recess of the injection element is clearly distinguishable, as the gray regions represent the walls of the injection element. At the most upstream location within the recess, red points are concentrated at the tip of the oxidizer injector (center) and the fuel injector (top and bottom). In contrast, the blue points at the most upstream flow in the recess correspond to the tip of the injector post wall, where mixing is already significant.

To verify the argument that "as the flow progresses downstream, fuel and oxidizer mix more extensively due to the increasing size of eddies driven by turbulence, resulting in greater dispersion of values from their mean," Figure 4 and Figure 6 can be examined. When the threshold for the variance $\widetilde{z''^2}$ is increased to 10^{-6}, more cells in the domain are highlighted in red, indicating enhanced mixing due to turbulent effects. Conversely, when the threshold for $\widetilde{z''^2}$ is decreased to 10^{-12}, fewer points are visible further downstream, as the stricter threshold reflects regions with lower mixing.

On the other hand, the argument "if $\widetilde{z''^2} = 0$, the mean mixture fraction $\widetilde{z}$ must be constant and can only take values of 0 or 1" can also be verified through DNS data for the values of $\widetilde{z''^2}$ non zero but very close to the zero. For the given condition $\widetilde{z''^2} < 10^{-8}$, with the help of an auxiliary script, there are 189796 cells for the one part of symmetric domain which is almost 6% of entire domain.



Figure 4: The illustration of cells where the variance $\widetilde{z''^2}$ is less than 10^{-6}. Red points highlight the locations of these cells.



Figure 5: The illustration of cells where the variance $\widetilde{z''^2}$ is less than 10^{-8}. Red points highlight the locations of these cells.



Figure 6: The illustration of cells where the variance $\widetilde{z''^2}$ is less than 10^{-12}. Red points highlight the locations of these cells.

In Figure 7, the y-axis represents the index number for each of the 189796 cells, while the x-axis represents the mean mixture fraction $\widetilde{z}$. As shown in the figure, when the variance $\widetilde{z''^2}$ is smaller than the threshold 10^{-8}, all values of the mean mixture fraction $\widetilde{z}$ are concentrated either very close to 0 or very close to 1. Therefore, it is theoretically meaningful to assume these values as either 0 (pure oxidizer) or 1 (pure fuel).

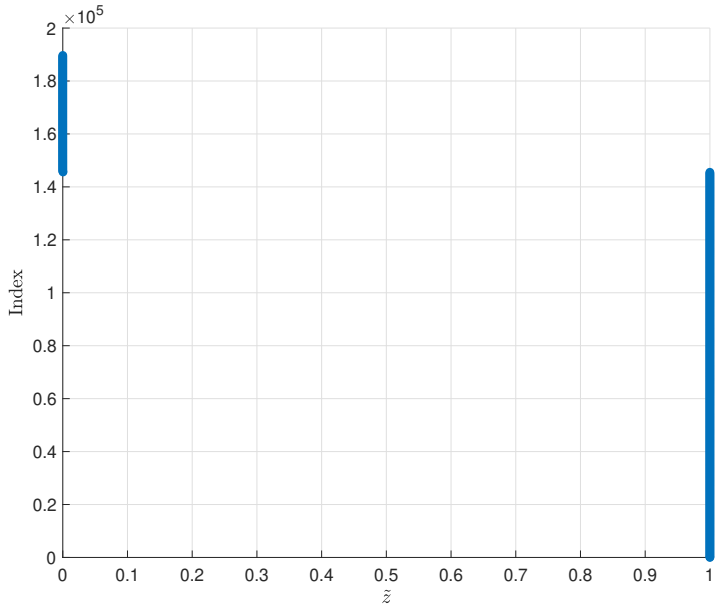


Figure 7: The illustration of the cells which have the value of variance $\widetilde{z''^2} < 10^{-8}$

The theoretical arguments are verified, and for the purposes of this study, it is adequate to set the threshold for $\widetilde{z''^2}$ at 10^{-8}. With this threshold established, the mean mixture fraction $\widetilde{z}$ values are assigned to be either 0 or 1 with 100% probability for these cells. For cells where $\widetilde{z''^2} > 10^{-8}$, the probability density function $p(z)$ is calculated using the β-PDF function. One critical consideration is the definition of the z-interval for constructing the probability density function for the remaining points. The z-interval must be defined as $0 < z < 1$ instead of $0 \leq z \leq 1$, as explained previously. In this study, the z-interval is defined between $0 + 10^{-5}$ and $1 - 10^{-5}$, discretized into 10,000 values.

This first script is capable of computing and graphically displaying the probability density function of the mixture fraction variable. The PDF distribution can be tested for different cells in the domain. For example, if a test cell is selected from the red area in Figure Figure 5, the variance $\widetilde{z''^2}$ is expected to be zero. Consequently, the mean mixture fraction $\widetilde{z}$ must be either 0 or 1 with 100% probability. This behavior is illustrated in the following figure:

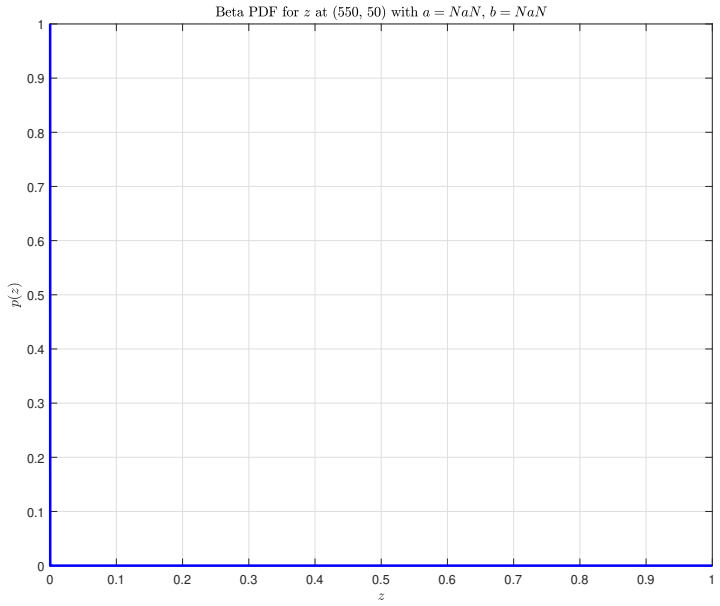


Figure 8: Probability density function of the mixture fraction z for a cell in the domain where the variance $\widetilde{z''^2} = 0$ and the mean mixture fraction $\widetilde{z} = 0$.

In Figure 8, an arbitrary cell in the domain with coordinates $x = 550$ and $z = 50$ is selected. At this cell, the variance $\widetilde{z''^2}$ is less than 10^{-8}, and thus, it is treated as $\widetilde{z''^2} = 0$. Additionally, since this cell is located in close vicinity of the oxygen injection element, the mean mixture fraction $\widetilde{z}$ is assumed to be constant with a value of 0.

Conversely, in Figure 9, an arbitrary cell in the domain with coordinates $x = 273$ and $z = 50$ is selected. This cell is located slightly above the previous cell and is in close vicinity of the fuel injection element. The variance $\widetilde{z''^2}$ remains less than 10^{-8}, but the mean mixture fraction $\widetilde{z}$ at this cell is constant with a value of 1.

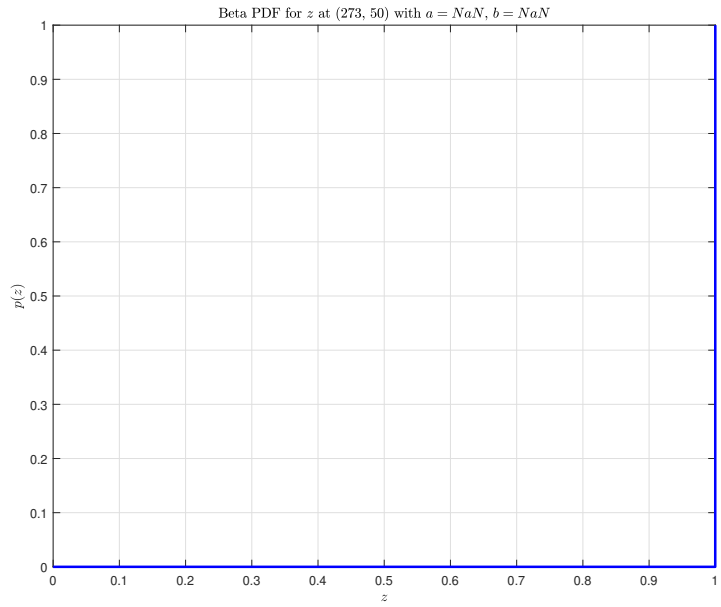


Figure 9: Probability density function of the mixture fraction z for a cell in the domain where the variance $\widetilde{z''^2} = 0$ and the mean mixture fraction $\widetilde{z} = 1$.

It can also be noted that the title of the figure includes the coordinates of the test cell along with the values of the constants a and b. For cells where $\widetilde{z''^2} < 10^{-8}$ (i.e., the points highlighted in red in Figure 5), it is not possible to calculate the values of a and b algebraically. As a result, these values are assigned as NaN.

Moreover, for a normal cell without any discrepancy, such as an infinitesimally small variance value (i.e., the points highlighted in blue in Figure 5), the expected probability density function is as follows:

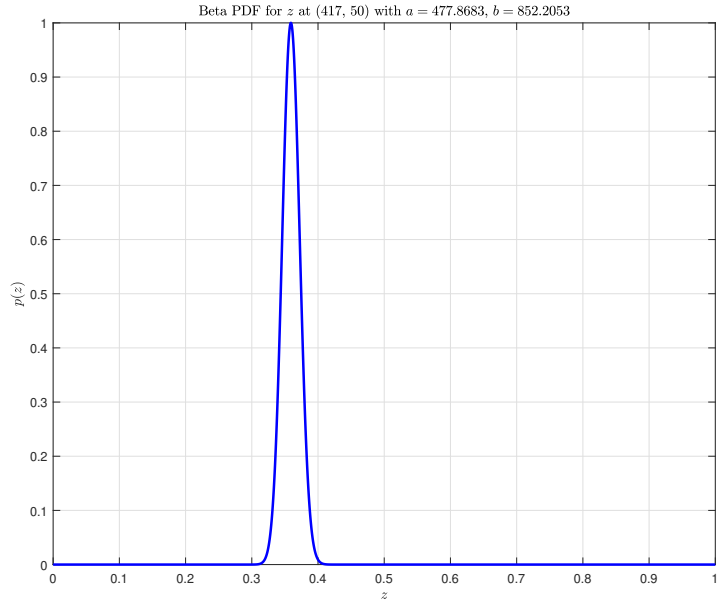


Figure 10: Probability density function of the mixture fraction z for a cell in the domain where the variance $\widetilde{z''^2} \neq 0$

The point in Figure 10 is located between the two previous points, sharing the same horizontal coordinate. The coordinates of this point are $x = 417$ and $z = 50$, and it is located on the injector post wall. The variance at this point is $\widetilde{z''^2} = 1.73 \times 10^{-4}$, which is larger than the presumed threshold

limit. This indicates that mixing effects at this point are significant, and the contribution of probabilities for different mixture fraction z values throughout the simulation must be considered. As shown in the figure, the mean mixture fraction is approximately $\widetilde{z} = 0.35$, while the probability density is distributed across mixture fraction z values ranging from approximately 0.3 to 0.4.

The next step is the prediction of the Favre-averaged temperature $\widetilde{T}$ for a cell. The script is designed to mathematically implement the Equation 25 ,which is exact an representation of the Equation 40. So far, the script has successfully constructed the probability density function $p(z)$ for a specific cell in the domain. Specifically, $p(z)$ has been computed for $10,000$ discrete z values within the range $0 + 10^{-5}$ to $1 - 10^{-5}$. The required values for the prediction of the Favre-averaged temperature $\widetilde{T}$ are provided by the flamelet library $T(z)$. In addition to the DNS results, the flamelet library from a previous experiment is also available. This flamelet library includes $10,001$ equally spaced z values ranging from 1.47×10^{-4} to 1.

An "if-else" loop is implemented in the script to predict the Favre-averaged temperature $\widetilde{T}$ for each cell. If the constants a or b are computed as NaN (i.e., the mixture fraction variance $\widetilde{z''^2}$ is smaller than the presumed threshold), a 100% probability is assigned to the cell being either pure fuel or pure oxidizer. If the mixture fraction variance exceeds the threshold, a standard probability density function is constructed (as shown in Figure 10). The Favre-averaged temperature $\widetilde{T}$ is then computed using Equation 25, as implemented in the script. For points located on the recess walls of the domain, the mean mixture fraction $\widetilde{z}$ is already assigned as NaN, and consequently, the Favre-averaged temperature $\widetilde{T}$ is also set to NaN.

To further illustrate, another random cell in the domain can be examined. The coordinates of this cell are $x = 395$ in the vertical direction and $z = 1365$ in the horizontal direction. The mean mixture fraction at this location is $\widetilde{z} = 0.34626$, and its variance is $\widetilde{z''^2} = 0.086748$. The probability density function for this point is plotted using MATLAB, as shown below:

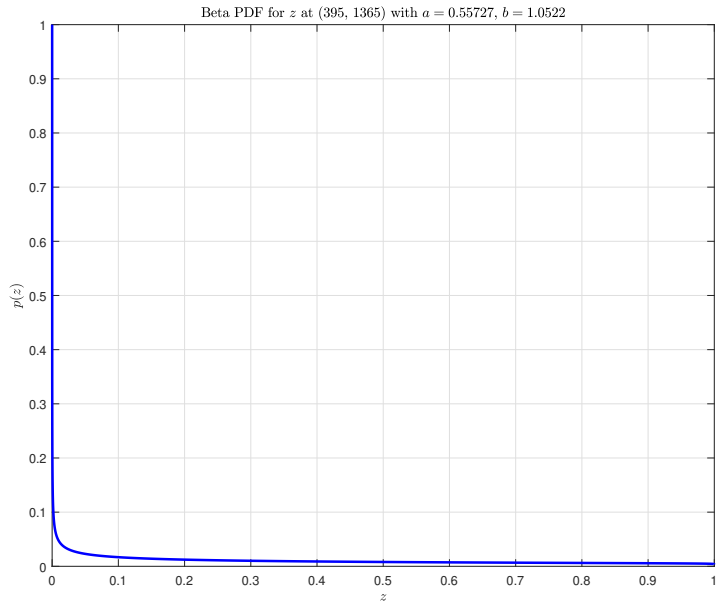


Figure 11: Probability density function of the mixture fraction z for given cell

The Favre-averaged temperature value for this cell is given as $\widetilde{T} = 2234.14\,\mathrm{K}$. On the other hand, using the aforementioned method, the computed Favre-averaged temperature for this cell is $\widetilde{T} = 1848.51\,\mathrm{K}$. In some cells, the accuracy of the predicted Favre-averaged temperature increases, while in others, it decreases. In certain cases, the predicted temperature matches the DNS results exactly. Therefore, the primary objective of this work is to predict the Favre-averaged temperature values using the aforementioned method with maximum accuracy, while also providing a theoretical justification for the discrepancies observed in cells with lower accuracy.

Second Step: Prediction of Favre-Averaged Temperature for Entire Domain

With the visualization of the mixture fraction PDF completed and the prediction of the Favre-averaged temperature for a single cell in the domain achieved using the first script, the next step is to utilize the second script to visualize the Favre-averaged temperature for every cell across the entire domain. The second script applies the same method described in the first step to each cell in the domain. Due to the axisymmetry in the domain, the computation is performed only for the 548×6232 cells within the domain.

However, the computation is time-consuming and inefficient, as the probability distribution function is calculated for $3,415,136$ distinct cells, each evaluated at $10,000$ discrete mixture fraction z values. To address this issue, a new method is introduced to perform computations on a coarser grid without sacrificing prediction accuracy. This method involves defining a new variable H to control the step size. When the computation begins for the first cell in the domain, the predicted Favre-averaged temperature $\widetilde{T}$ value for that cell is assigned to the next H cells in both the vertical and horizontal directions. Subsequently, the computation resumes at the $(H+1)^{\text{th}}$ cell in both directions, and the process repeats. The effect of step size value H to the computational time can be visualized as:

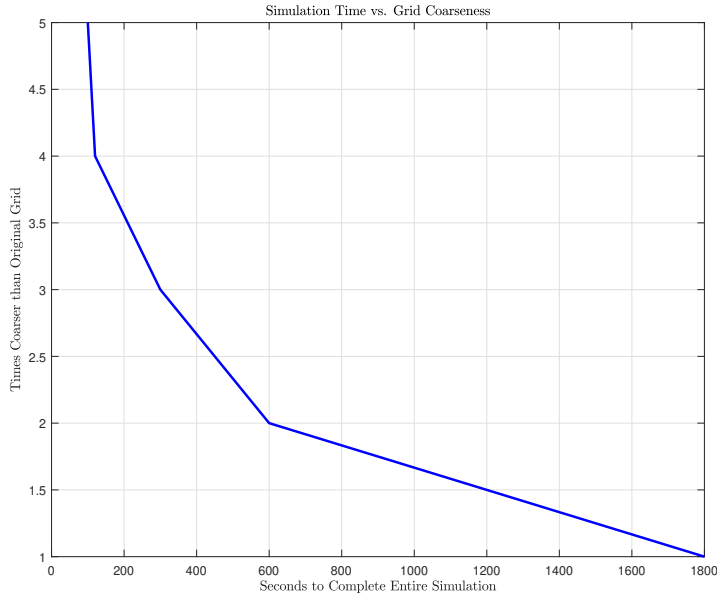


Figure 12: Effect of Step Size H on Computation Time

The computation time has been recorded for five different values of H, and the relationship between H and computation time is visualized in Figure 12. As shown in the figure, even setting $H = 2$ reduces the computation time by approximately 66%. Therefore, for validation purposes, the step-size method is occasionally employed to significantly decrease computation times across the domain.

Nevertheless, for demonstration purposes, H is set to 1, allowing the Favre-averaged temperature $\widetilde{T}$ prediction to be visualized for the entire domain, as shown below:

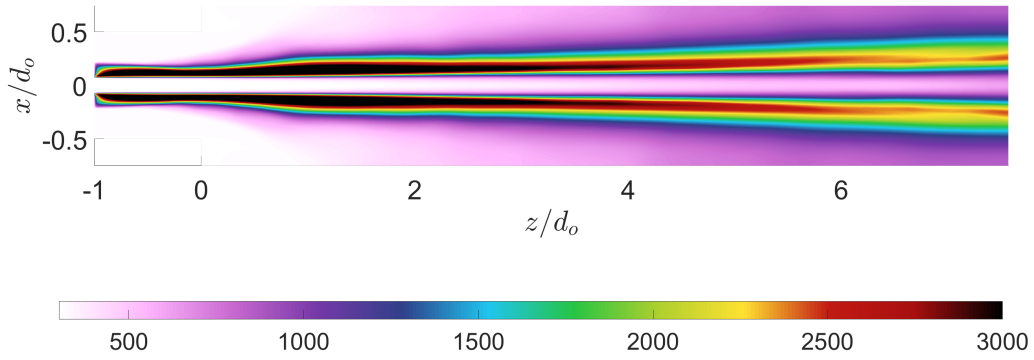


Figure 13: Favre-averaged temperature $\widetilde{T}$ distribution across the domain, computed with a step size of $H = 1$.

The figure above represents the initial attempts at predicting the Favre-averaged temperature distribution across the entire domain. With the DNS-derived Favre-averaged temperature results available, the goal is to compare the two temperature distributions (predicted vs. exact) and improve the accuracy of the predictions by refining either the construction of the probability density function or the resolution of the mixture fraction interval. The comparison between predicted and exact Favre-averaged temperature values will be made on the Results and Discussion section.

Third Step: Introduction of Skewness (γ)

As an improvement, the effect of skewness on the construction of the probability density function can be investigated. As previously mentioned, the β-function-based PDF is constructed using two variables: the mean and variance of the mixture fraction. However, incorporating a higher-order moment, such as skewness, may improve the accuracy of the PDF. Skewness is a statistical measure that quantifies the asymmetry of the probability distribution of a real-valued random variable about its mean. While variance measures the spread of values around the mean, skewness provides additional information about the distribution's shape, which could lead to better results for certain locations.

Typically, higher-order moments of the mixture fraction, such as skewness, can be obtained by solving Equation 16, similar to how the variance is computed. An additional equation is derived and closed to calculate the variance. However, for the purposes of this study, to minimize the computational workload, skewness values are directly obtained from the DNS-based probability density function results. With the inclusion of skewness, the probability density function must now be constructed in terms of the mean mixture fraction $\widetilde{z}$ and skewness, once again expressed through the constants a and b. However, the challenge lies in explicitly expressing a and b as functions of the mean and skewness.

According to [12], the skewness of the beta distribution can be expressed as:

$$\widetilde{z''^3} = \frac{2(b-a)\sqrt{a+b+1}}{(a+b+2)\sqrt{ab}}, \tag{43}$$

where a and b are the shape parameters of the beta distribution. Additionally, the mean value of the distribution is given by:

$$\widetilde{z} = \frac{a}{a+b}. \tag{44}$$

The challenge lies in the algebraic complexity of explicitly expressing a and b in terms of skewness $\widetilde{z''^3}$ and mean $\widetilde{z}$, as was done in Equation 38 and Equation 39 for mean and variance. Therefore, the values of a and b must be determined by numerically solving Equation 43 and Equation 44 using an auxiliary MATLAB script.

At each target point in the domain, the corresponding values of the mean ($\widetilde{z}$), variance ($\widetilde{z''^2}$), and skewness ($\widetilde{z''^3}$) are extracted from the provided DNS dataset. The primary goal of the MATLAB script is to explicitly compute the parameters a and b of the Beta distribution by solving the nonlinear Equation 43 and Equation 44, which relate these parameters to the statistical moments of the distribution.

To avoid confusion with the final a and b matrices that store the complete solution over the entire field, the intermediate values calculated during the estimation process are referred to as a_2 and b_2. Once a_2 and b_2 are determined at a given point, the corresponding Beta probability density function (PDF) is fully specified. This procedure is repeated for every point in the 548×6232 spatial domain, leveraging the statistical symmetry across the dataset.

However, certain degenerate cases must be handled separately. Specifically, when the variance is extremely small (i.e., below a predefined threshold), or when the skewness value is missing or deemed unreliable (e.g., `NaN`), the standard estimation procedure for calculating the Beta distribution parameters becomes invalid. In such cases, the probability density function (PDF) is computed solely based on the available mean and variance values, while the corresponding PDF associated with the skewness-based estimation is explicitly set to `NaN` to indicate its invalidity.

If, in addition to missing skewness, the variance is also negligible, the already mentioned heuristic approach is employed to assign the Beta distribution parameters. This heuristic is based on the proximity of the mean value to the bounds of the distribution: if the mean is closer to zero, the distribution is assumed to be fully oxidizer-rich and the parameters are set as $a = 0$, $b = 1$; conversely, if the mean is closer to one, the mixture is considered fully fuel-rich, and the parameters are assigned as $a = 1$, $b = 0$. This strategy ensures that every point in the domain is assigned a consistent and interpretable distribution, even in the presence of degenerate or incomplete data for skewness and mean based probability density function.

Finally, this auxiliary MATLAB script computes the a_2 and b_2 values for the valid grid points and stores them in a file named `ab_mean_skew.mat`, which is used in subsequent post-processing scripts. The two nonlinear equations that must be solved are given by:

$$F_1(a,b) = \widetilde{z} - \frac{a}{a+b} \tag{45}$$

$$F_2(a,b) = \widetilde{z''^3} - \frac{2(b-a)\sqrt{a+b+1}}{(a+b+2)\sqrt{ab}} \left(\widetilde{z''^2}\right)^{3/2} \tag{46}$$

These are solved numerically using MATLAB's `fsolve` function. Because `fsolve` is an iterative nonlinear solver, the degree of convergence of the solution heavily depends on the initial guess quality. In order to become more efficient and solid, the iterations begin with the (a, b) values computed from the same grid point solely on the basis of the mean and the variance, as these provide a fair starting point for the optimization process. The residuals for the estimated values of a_2 and b_2 are bounded to fall

below a tolerance of 10^{-4}, ensuring the accuracy of computed parameters.

Calculation of β PDFs in terms of $\widetilde{z}$ and $\widetilde{z''^3}$

Since the a_2 and b_2 values are computed for the valid grid points and stored as matrix, it is time to calculation of β PDFs in terms of $\widetilde{z}$ and $\widetilde{z''^3}$ as it is already done for $\widetilde{z}$ and $\widetilde{z''^2}$. The aim of doing that is to asses whether β PDFs in terms of $\widetilde{z}$ and $\widetilde{z''^3}$ could be used instead of $\widetilde{z}$ and $\widetilde{z''^2}$ in order to predict the Favre-averaged temperature $\widetilde{T}$ across the domain. This step also requires an auxiliry MATLAB script to work functionally.

The algorithm in this MATLAB script utilizes the previously computed a_2 and b_2 values. For each grid point in the symmetric 548×6232 spatial domain, the script calculates the Probability Density Function (PDF) using those a_2 and b_2 values, in a similar way to how the PDF was calculated based on the mean and variance previously. However, there are some key nuances to be considered.

Firstly, if the mean value is NaN (which locates the recess walls), the corresponding PDF value is also set to NaN, indicating that no valid PDF can be computed for that grid point. Similarly, in the case of a degenerate situation where the variance value is below the threshold, the a_2 and b_2 values are already computed as NaN which means there is no valid PDF for those points as well. As a result, the PDF value in terms of both mean and skewness will also return NaN in such cases.

Moreover, up until this point, the PDF in terms of mean and variance was constructed for 10,000 discrete z values between 0 and 1, which effectively assesses the probability for a wide range of z-values at each grid point. However, the approach now has to be shifted further. The PDFs will now be constructed using only 51 specific discrete z values between 0 and 1. This change allows for a more focused analysis of the constructed PDFs, which enables a comparison between the newly generated PDFs and the actual one derived from the previously mentioned DNS results. These real PDF values are retrieved from the transient DNS, which is built by using a sufficiently high number of realizations.

To summarize, the script now calculates the Beta PDFs in terms of both mean and variance, as well as mean and skewness, for 51 discrete z-values. The reason behind that is to assess the accuracy of the models by comparing the constructed PDFs against DNS results based PDFs. Before explaining the comparison methodology, it is cruical to provide better understanding of the real PDFs and the way that they are constructed from the transient DNS data.

In Figure 14, the probability density function of the mixture fraction z based on mean & skewness is shown to differ slightly from the probability density function constructed using mean & variance. Since both PDFs are conditioned on the same mean value, they share equivalent mean values. However, noticeable qualitative differences arise between the distributions, especially near the extremes. Namely, the vicinity around pure oxidizer (z=0) and pure fuel (z=1). A further detailed quantitative assessment of these discrepancies will be conducted and analyzed in the following sections.

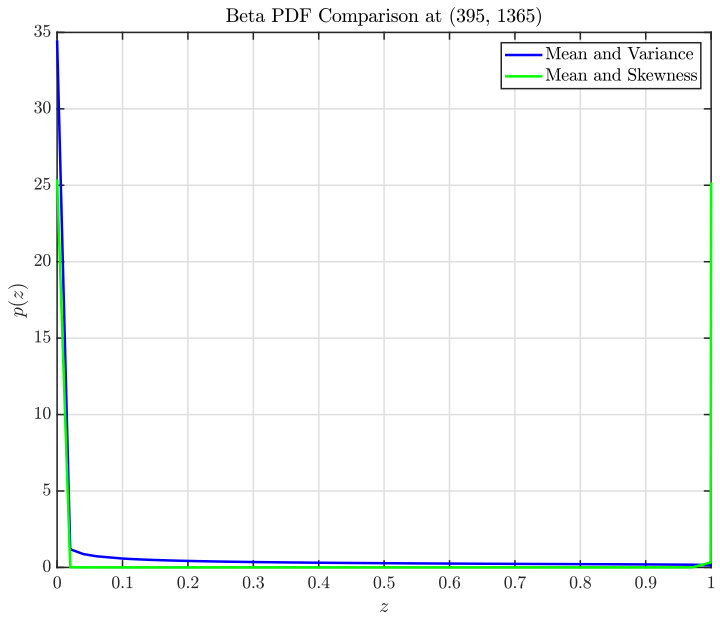


Figure 14: Probability density function of the mixture fraction z in terms of both mean & skewness and mean & variance for given cell

Real PDF Based on Probability of 51 Discrete Mixture Fraction Values

As previously mentioned, all data used in this study were obtained from a comprehensive Direct Numerical Simulation (DNS). The DNS was executed for a sufficiently long time to ensure that all transient effects were smoothed out, combustion fully developed, and time-dependent changes in flow variables reached a statistically steady state. Although the simulation is run on high-performance computing clusters comprising hundreds of processors, the DNS required several days to complete due to its computational intensity. The simulation was performed transiently for $t > 0$ seconds, and all relevant flow quantities were subsequently Favre-averaged over the entire period.

In the previous section, it was mentioned that the latest probability density functions (PDFs)—constructed in terms of both mean and variance, and mean and skewness—were generated using only 51 discrete mixture fraction z values between 0 and 1. This discretization was chosen to enable direct comparison between the modeled PDFs and the actual distributions obtained from transient DNS results. The comparison procedure was carried out as follows:

1. For the entire 548×6232 axisymmetric domain, probes were assigned to each individual cell.
2. At each time step throughout the simulation, the local mixture fraction z value was recorded for every cell.
3. The frequency of each realized mixture fraction value within the range $[0, 1]$, discretized into 51 intervals, was compiled into a matrix. Corresponding probability values (based on the number of realizations) were recorded in a separate matrix.
4. As a result, two distinct 3D matrices were constructed: one representing the realized mixture fraction z values across all cells, and the other capturing the corresponding probability values.

Notes on the Realization Measurement:

- Ideally, the 51 discrete z values should be uniformly spaced with an interval of 0.02. However, due to limitations in the number of measurements available within the probe cells, it was not always feasible to obtain values that are exact multiples of 0.02. To mitigate this issue, the nearest available value to each ideal z point was used in place of the exact target value.

- In cases where no realizations were observed within a specific interval (e.g., realizations exist at 0.40 and 0.44 but none between), a value of `NaN` was automatically assigned to the corresponding row of the realization matrix, while a value of 0 was assigned to the equivalent row in the probability matrix. However, for error calculations (to be discussed in the following chapter), these `NaN` entries are replaced with the values from a default z matrix consisting of 51 uniformly spaced rows. This default matrix represents the ideal z structure with a consistent spacing of 0.02.

To provide a clearer understanding of the data structure, these measurements can be visualized in matrix form. As an illustrative example, the probe cell located at coordinates (395, 1365) has been selected to examine its corresponding realization matrix and associated probability matrix. The original matrix, the default z matrix, and the modified (replaced) matrix are presented in Appendix A.

Error Prediction and Combined β PDF Construction

The β PDFs based on mean and variance, as well as those based on mean and skewness, were derived in the previous section. Additionally, the concept of a real β PDF has been introduced, which allows for quantifying the difference between these two formulations. The objective of this section is to evaluate all cells to determine whether either form of the β PDF offers improved accuracy in predicting the probability distribution for a given cell. Depending on the cell, either the β PDF defined by mean and variance or the one defined by mean and skewness may yield better predictive performance. However, the analysis is not restricted to selecting one over the other; instead, the contributions of both types of β PDFs will be considered collectively. This evaluation will be performed using an error prediction algorithm.

$$p_{\text{comb}}(z) = c \cdot p_{\text{mean_var}}(z) + (1 - c) \cdot p_{\text{mean_skew}}(z) \tag{47}$$

$$\text{error} = \int_0^1 |p_{\text{real}}(z) - p_{\text{comb}}(z)| \, dz \tag{48}$$

$p_{\text{comb}}(z)$ represents the newly constructed **Combined Probability Density Function**, incorporating contributions from both $p_{\text{mean_var}}(z)$ and $p_{\text{mean_skew}}(z)$. The weight factor c, shown in Equation 47, determines the relative influence of each component. This combined PDF will be utilized for the final Favre-averaged temperature prediction across the computational domain.

Once $p_{\text{comb}}(z)$ is constructed, the error between the real PDF and the combined PDF will be calculated for each value of the mixture fraction z. This process will be repeated at every point in the 548×6232 spatial domain. The primary objective for each cell is to identify the optimal value of c (c_{opt}) that minimizes the error defined in Equation 48.

The optimal c value will be computed using an auxiliary MATLAB script that employs the `fminbnd` function to minimize the error. Additionally, the integral in Equation 48 is numerically evaluated using MATLAB's `trapz` function.

Effect of Modified z Matrix on Error Calculation

As discussed in the previous chapter, the original z matrix was modified (`NaN` values were replaced with entries from the default z matrix) in order to facilitate accurate error calculations. The reason behind this modification will now be illustrated using a representative probe point.

Both Figure 15 and Figure 16 present Beta PDF comparisons for the probe point (395, 1365). This is the same point for which both the modified and unmodified z values, along with their corresponding

probabilities, are provided in Appendix A. As shown in the unmodified version (Figure 15), for the real PDF, the first few entries between 0 and 0.1 in the z matrix contain `NaN` values. If these values had not been replaced with the appropriate values from the default z matrix, the error expression in Equation 48 would have failed to evaluate discrepancies in these early entries. This would have led to inaccurate error calculations and, consequently, an incorrect estimation of the optimal weight factor c.

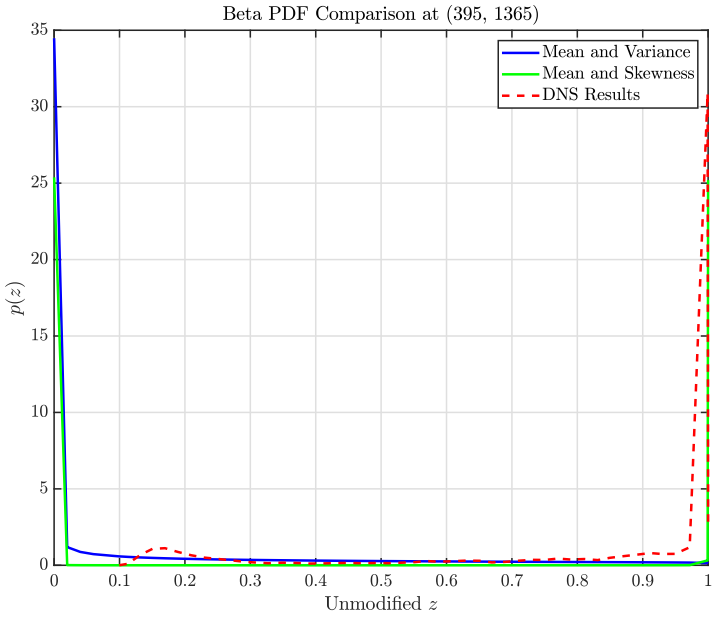


Figure 15: Distribution of PDFs with unmodified z matrix at the location (395,1365)

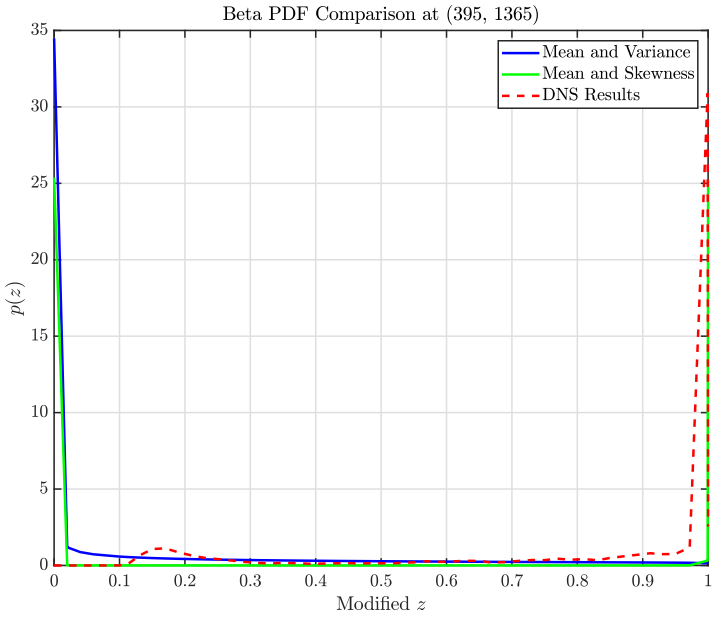


Figure 16: Distribution of PDFs with modified z matrix at the location (395,1365)

Favre-Averaged Temperature $\widetilde{T}$ Prediction Using the Combined β PDF

For the entire 548×6232 axisymmetric domain, optimal c values have been computed and stored. This procedure was initially performed using 51 discrete z values, as the real PDF data is based on this resolution, as previously discussed. To enhance prediction accuracy, the combined PDF will now be reconstructed across the full domain using a denser discretization. Specifically, the z-interval from

$0+10^{-5}$ to $1-10^{-5}$ will be divided into 10,000 discrete values. This finer resolution enables improved error estimation and a smoother representation of the probability distribution, due to the increased number of z samples.

The finalized probability density function (PDF) will be used to predict the Favre-averaged temperature $\widetilde{T}$, across the entire computational domain. The temperature distribution will be computed following the same methodology previously mentioned in Equation 40. Throughout this process, the local flamelet assumptions in Figure 3 will remain valid and continue to serve as the basis for accurate modeling. All temperature distribution plots, along with comparative analyses of temperature differences, will be presented in the Results and Discussion chapter. These visualizations will provide insight into the accuracy and effectiveness of the combined β PDF approach in predicting Favre-averaged temperature across the domain.

Furthermore, the potential correlation between z and c_{opt} will be examined. It is expected that c_{opt} exhibits a relationship with at least one flow variable or combustion-related parameter. For instance, in near-wall regions, the contribution from the PDF based on mean and skewness is observed to be nearly zero. As another example, in the vicinity of the flame front, the optimal c value typically lies within the range of 0.4 to 0.6. A more detailed investigation into these correlations, along with identification of relevant parameters, will be presented in the Results and Discussion chapter.

3 Results and Discussion

3.1 Local Evaluation of Predicted PDFs

In order to better understand the calculations and plots made for the whole domain, firstly a local point-wise evaluation is conducted. For this purpose, PDF predicted by mean&variance (classical distribution or $p_{\text{mean_var}}$) and mean&skewness (modified distribution or $p_{\text{mean_skew}}$) are compared locally at three locations that vary radially in the combustion chamber. The three points are shown in Figure 17.



Figure 17: Locations of the three points of evaluation shown in the combustion chamber. The color visualizes the mixture ratio: blue for $Z = 0$ (oxidizer) and red $Z = 1$ (fuel)

First location at $x = 208mm$ Figure 18, represents a point in the chamber where the unmixedness is quite high, with a large portion made up of pure hydrogen. The smaller portion is at $z = 0,2$, which is mainly oxygen. This peak has a higher variance than the largest peak. At this location, the $p_{\text{mean_var}}$ ignores the oxygen-rich region peak and predicts the point as containing just fuel. Which is a reasonable prediction since the highest probable z value is 1. On the other hand, $p_{\text{mean_skew}}$ also ignores the oxygen-rich peak while also having the fuel-rich peak not exactly at $z = 1$. Even though not capturing all the effects in the real PDF perfectly, $p_{\text{mean_var}}$ performs overall better at this location. However, at z-values larger than 0,7 a combination of both methods can prove beneficial since the real values fall in between. While the $p_{\text{mean_skew}}$ over-predicts, the $p_{\text{mean_var}}$ d under-predicts at nearly 0.

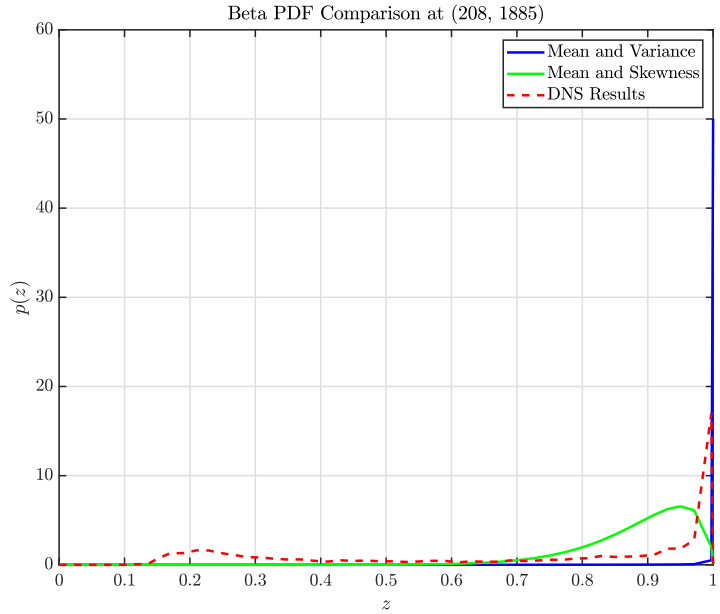


Figure 18: Comparison of predicted PDFs with the DNS PDF at the location 1 (208,1855)

Moving about a $150mm$ in the $-y$ direction Figure 19, a similar real PDF distribution is seen, with the only difference being an even more probable occurrence of $z = 1$. This time, $p_{\text{mean_var}}$ also predicts oxygen in the flow and puts its probability as being almost as probable as the hydrogen. The peak at the oxygen-rich point at $z = 0,2$ seems to cause the beta distribution to put another probability peak at $z = 0$. $p_{\text{mean_skew}}$ on the other hand, puts its peak probability almost at the middle $z = 0,5$ and

thus predicts a well-mixed gas composition at this location. A potential improvement by the inclusion of skewness does not seem possible for this location.

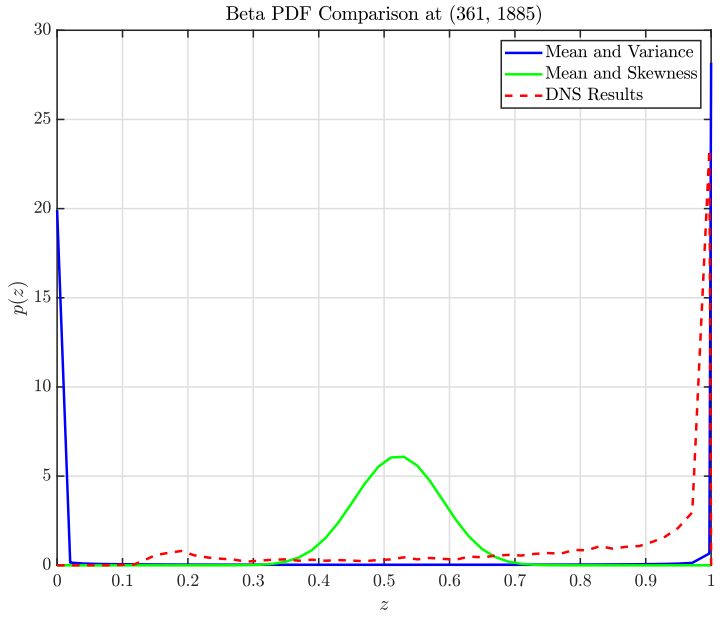


Figure 19: Comparison of predicted PDFs with the DNS PDF at the location 2 (301,1855)

As the last point further in the core region Figure 20, a quite large difference in PDF prediction behavior can be observed. The PDF from DNS results shows a similar distribution to the one before. The highest probability peak is located at $z = 1$, while a smaller peak, although significantly higher than the previous location, appears at the highly oxygen-rich z value of approximately $0,05$. Both predictions made at this location are exactly the same. Interestingly, beta distributions foresee a location where only pure oxygen appears. Considering that real PDF peaks at $z = 1$ and having only a small portion of oxygen, predictions made by both distributions are quite inaccurate.

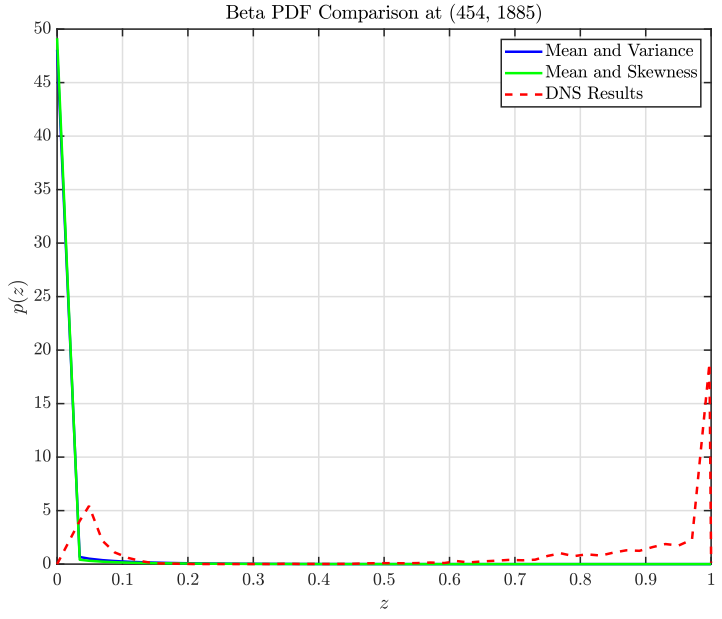


Figure 20: Comparison of predicted PDFs with the DNS PDF at the location 3 (454,1855)

3.2 Local Evaluation of Combined PDFs

Once the $p_{\text{mean_var}}$ and $p_{\text{mean_skew}}$ are evaluated, a further local analysis is done for the same points. This time also includes the combined PDF. As explained in Application of Method, the combined PDF $p_{\text{mean_comb}}$ is made up of both the $p_{\text{mean_var}}$ and the $p_{\text{mean_skew}}$ weighted by the coefficient c.

In Figure 21 $p_{\text{mean_comb}}$ is evaluated at the point 1. Although with a c value of $0,7708$, which is a quite high skewness contribution, the $p_{\text{mean_comb}}$ seems to fall completely in line with the mean&variance. This is the case even in the previously highlighted interval of $z = 0,7 - 1$, where skewness has a high potential of increasing the accuracy of $p_{\text{mean_var}}$ by pushing its values to higher probabilities, thus closer to DNS results. This behavior can be explained by the high accuracy of the $p_{\text{mean_var}}$ over the whole domain. The relatively higher error in the region $z = 0,7 - 1$ makes a small impact on the calculation of c thus the overall distribution of the $p_{\text{mean_comb}}$.

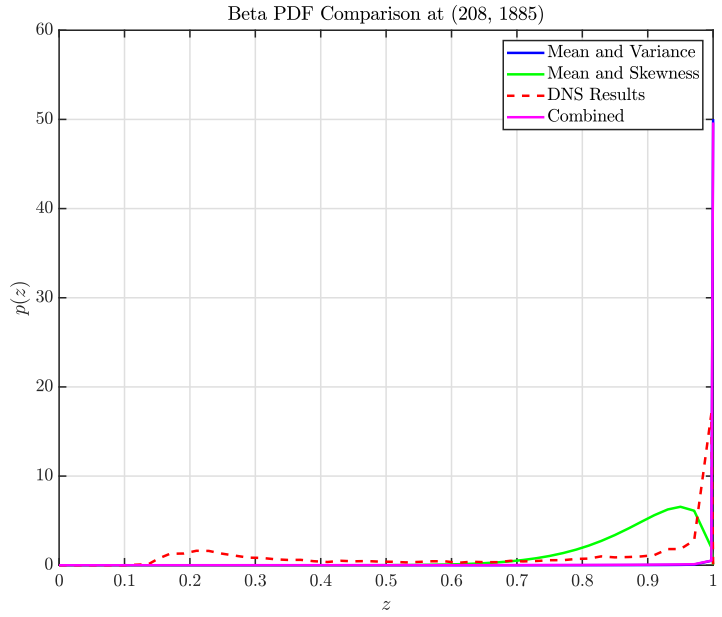


Figure 21: Comparison of DNS, Moment Based, and Combined PDF with $c = 0.7708$ at the location 1 (208,1855)

The distributions in the two points of the core region of the combustion chamber are shown in Figure 22 and Figure 23. The combined PDFs for these points are calculated with a c value of almost 1.

The prior point is quite interesting for evaluating the combined PDF since it showcases two very different distributions of classic and modified PDF. At this position, the combined PDF is even closer to the classic beta distribution. The difference between the distributions lie in the interval $z = 0,35 - 0,7$ with modified distribution majorly over predicting probability in this region. This region alone is responsible for reducing the c value minimally thus causing the combined PDF distribution to be marginally higher than the standard distribution. For this position in the chamber, combined PDF fails to capture the real distribution.

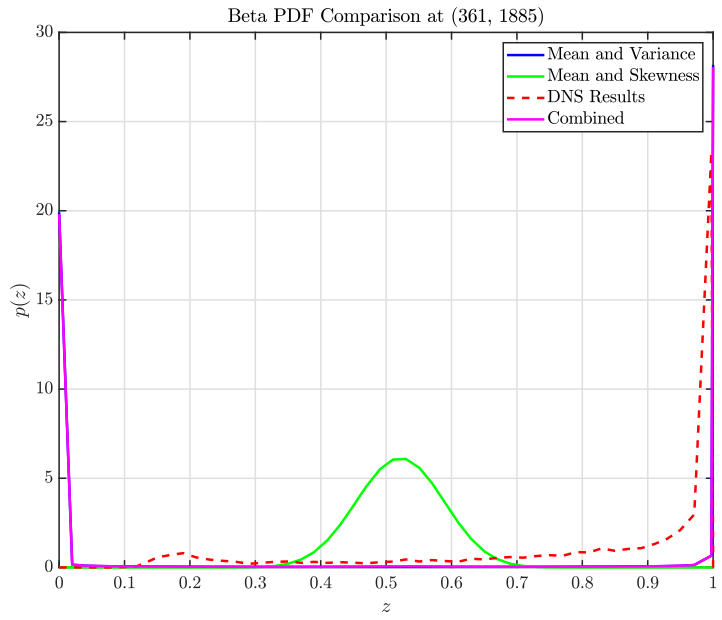


Figure 22: Comparison of DNS, Moment Based, and Combined PDF with $c = 0.9399$ at the location 2 (361,1855)

The latter point positioned around the oxidizer side of the stoichiometric mixture ratio is characterized by strong fluctuations between the oxidizer and fuel flow. As mentioned previously, all the distributions, among them also the combined PDF, fails to capture this phenomenon and all three of them almost fall in line. The combined PDF also puts the point completely in the oxidizer flow and fails to represent the major spike in $z = 1$.

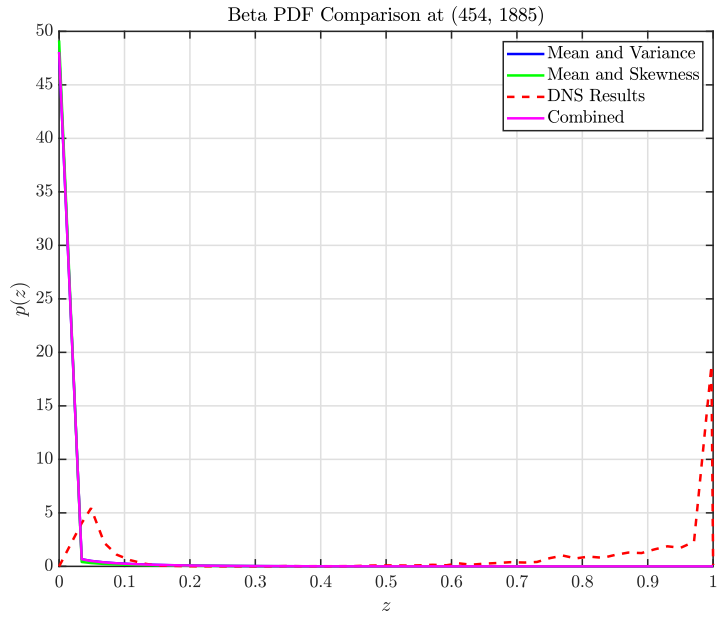


Figure 23: Comparison of DNS, Moment Based, and Combined PDF with $c = 0.9999$ at the location 3 (454,1855)

3.3 Prediction of the Temperature Field

After analyzing the combined PDF at three locations within the combustion chamber, an assessment of its predictive performance over the entire domain was carried out. To this end, the temperature field was calculated using the mixture fraction PDFs, and the resulting distributions were compared against reference DNS data. This comparison highlights the discrepancies between the predicted and actual temperatures, thereby allowing a critical evaluation of the combined PDF's accuracy in capturing temperature.

Furthermore, the coefficient c, was examined in detail. Its spatial variation across the combustion chamber was visualized. Based on the observed trends, possible physical explanations were proposed for the local changes in c.

Figure 24 and Figure 25 visualize the temperature field predicted by the PDF constructed based on mean and variance ($p_{\text{mean_var}}$), and the mean and skewness ($p_{\text{mean_skew}}$), respectively. Figure 24 features a smooth field with maximum temperatures occurring at and around the stoichiometric ratio. Figure 25, on the other hand, shows sharp gradients at the boundaries of the maximum temperature region. From the maximum temperature region of around $3000K$, a sharp decline to around $700K$ is observed. These sudden changes are also very prominent at the beginning of the domain, where propellants are introduced. Injection temperature seems to carve out pockets of high temperature. Additionally, the temperature distribution is not as smooth as that shown in Figure Figure 24. This discrepancy arises because the mean and variance values are directly obtained from the DNS results, while the skewness is computed indirectly using the mean and variance of each cell, as described in Equation 43.

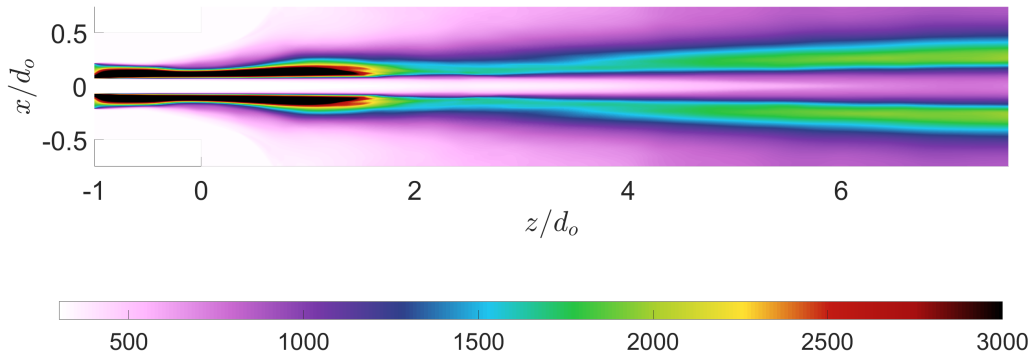


Figure 24: Prediction of $\widetilde{T}$ Based Solely on $p_{\text{mean_var}}$

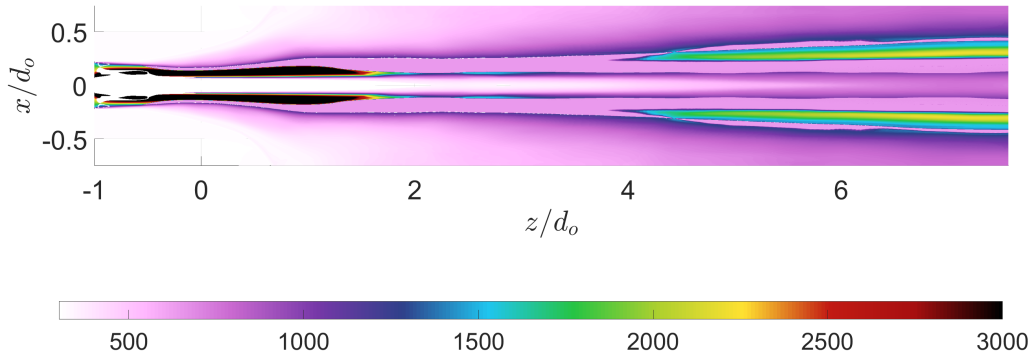


Figure 25: Prediction of $\widetilde{T}$ Based Solely on $p_{\text{mean_skew}}$

In plots Figure 26 and Figure 27 the difference between previous plots and the DNS data are visualized. This difference quantifies the error of prediction. Biggest errors are situated at and around the stoichiometric mixture fraction. In this region both distributions under predict the flame temperature by $300K$. For the skewness distribution there exists a region at the last third of the domain where sharp gradients in error become present. A streak of low error region is also present here.

While in the rich mixture conditions (outer region of the domain) the standard beta distribution has a tendency to over predict, the skewness distribution under predicts the temperature. Thus has the combined PDF the potential to increase accuracy in this region. Because the standard beta distribution goes from under prediction to over prediction while moving from the stoichiometric to fuel rich regions, there is a horizontal line of almost zero error. This region does not exist for the skewness distribution.

Overall perform both distributions quite similarly and the combined PDF has the potential to increase the prediction accuracy on a broad region, especially in the rich outer parts.

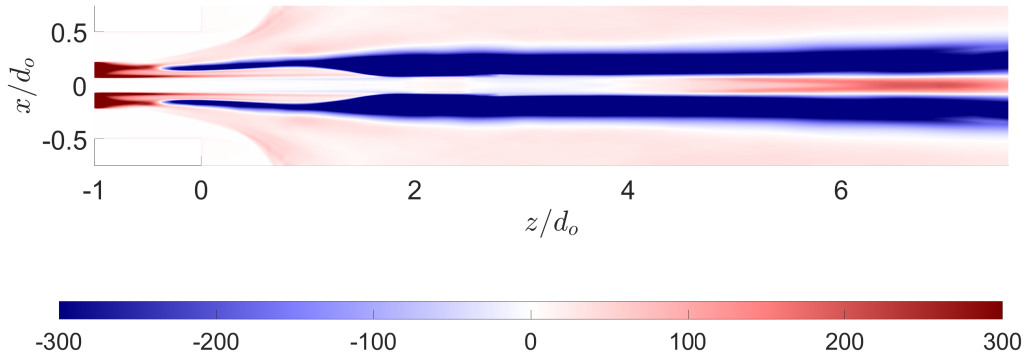


Figure 26: $\widetilde{T}_{\text{meanvar}}$ - $\widetilde{T}_{\text{DNS}}$

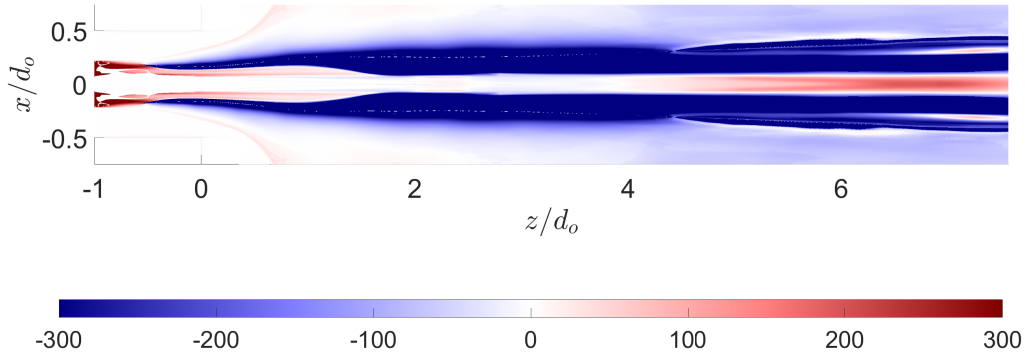


Figure 27: $\widetilde{T}_{\text{meanskew}}$ - $\widetilde{T}_{\text{DNS}}$

After assessing the accuracy of temperature predictions using both the standard Beta PDF and the distribution incorporating mean&skewness, and having identified specific regions with potential for improvement, the combined PDF model is subsequently evaluated.

In Figure 28 the predicted temperature field can be seen. Characteristic sudden temperature changes around the stoichiometric mixture that were prominent in the field predicted by skewness distribution are also present in combined PDF. In the outer region of the flame there exists a pocket of higher temperature marked by blue ($1500K$. This seems to be caused by a higher portion of standard beta distribution as opposed to surrounding lower temperature region, which is a feature of skewness distribution. The field downstream of the combustion chamber is almost exactly the same as mean&variance field.

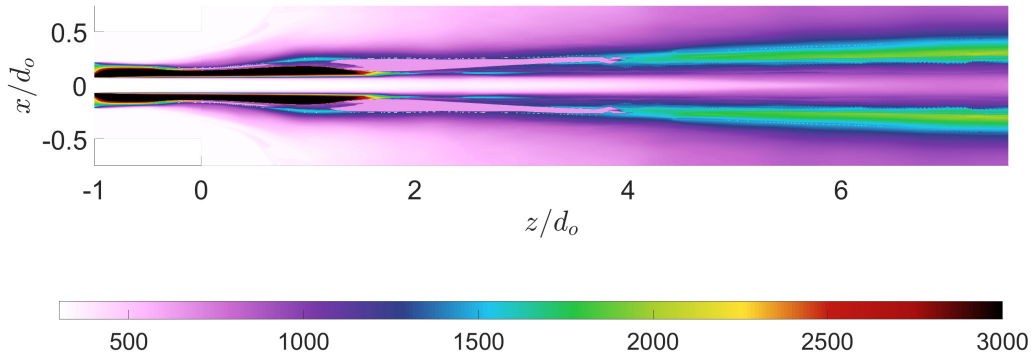


Figure 28: Prediction of $\widetilde{T}$ Based on p_{comb}

In Figure 29 the prediction error of the combined PDF can be compared with error of the beta distribution. In the outer region of the domain over prediction error, although still present, is noticeably reduced. The under prediction region around the stoichiometric ratio grew in the middle domain. The downstream features the characteristic streaks of near-zero error region just like in the temperature field calculated by mean&skewness distribution.

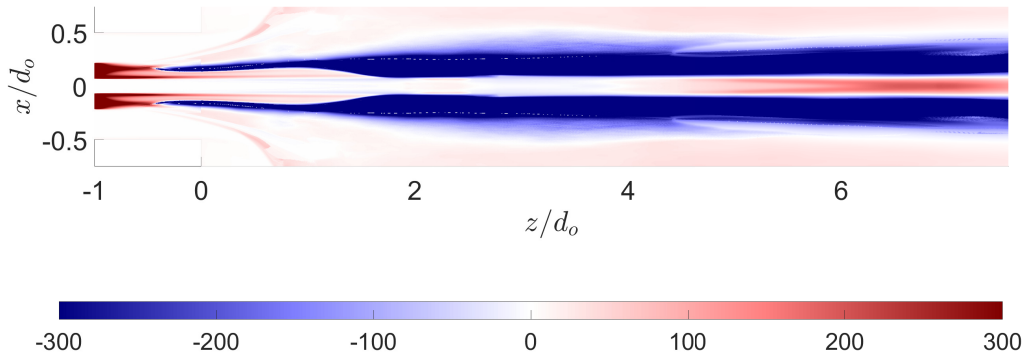


Figure 29: $\widetilde{T}_{\text{comb}}$ - $\widetilde{T}_{\text{DNS}}$

Figure 30 shows how the optimal c value was calculated for each point in the domain. The most outer parts of the combustion chamber which are characterized by rich mixture fractions, almost exclusively use the classic beta distribution. While moving from the outer region to the flame skewness contribution increases. This was also apparent in the difference plot as the under prediction region was larger in comparison with the beta distribution prediction. There are even regions who rely solely on mean&skewness marked by white in the plot. Overall the field is quite irregular with large changes happening between close regions. The very core of the plot features a noisy region where numerical errors are present.

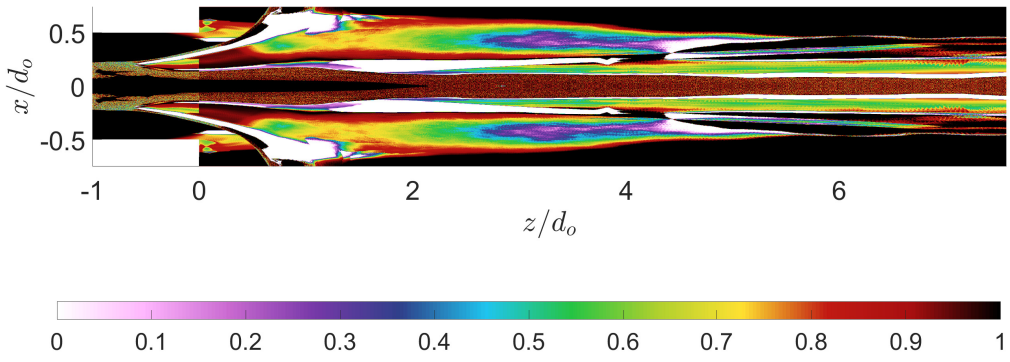


Figure 30: Spatial distribution of optimal c values across the computational domain

4 Conclusion and Future Work

The figures presented in Section Results and Discussion clearly demonstrate that incorporating higher-order moments of the mixture fraction significantly enhances the accuracy of Favre-averaged temperature predictions throughout the domain. However, several methodological questions remain unresolved. For instance, it is yet to be determined whether the additional computational effort justifies the gains in predictive accuracy. Additionally, the existence of a global correlation between the optimal c value and specific flow characteristics or simulation parameters remains an open question—such a correlation could potentially be leveraged in broader simulation frameworks. Furthermore, there are instances where the inclusion of skewness may lead to reduced predictive accuracy in localized regions. The following sections aim to investigate these considerations using the results obtained from the conducted simulations.

First of all, Figure 26 and Figure 29 demonstrate that improvements in prediction accuracy are primarily localized near the inlet region. In contrast, farther downstream, the predicted Favre-averaged temperature either remains unchanged or exhibits only negligible variation. From this perspective, the inclusion of skewness might appear unnecessary, given the computational effort required. Nevertheless, as emphasized in the objective of this study, even small improvements in accuracy are valuable and warrant consideration.

One underlying factor is that the optimal c values were derived based on the real PDF data (Application of Method), which consists of only 51 discrete z values. However, to gain a more global and robust understanding of the performance of PDFs constructed from mean–variance $p_{\text{mean_var}}$ and mean–skewness $p_{\text{mean_skew}}$ formulations, further investigation is necessary using a larger number of datasets. In practical simulations, access to such real PDF data (obtained via direct numerical simulation) is generally not available. For each spatial cell within the combustion chamber, a thorough understanding is needed of the conditions under which the β PDF constructed using mean and skewness more accurately replicates the behavior of the real PDF. Identifying these regions and the underlying flow or combustion characteristics that drive this alignment is crucial for improving predictive models. As part of future work, this relationship can be further investigated by employing a higher number of realizations and a more finely discretized set of z values, which would provide a more detailed and statistically robust basis for analysis.

When the aforementioned procedure is repeated with a larger dataset, optimal c values should once again be computed for the entire spatial domain. In this study, a potential correlation between the optimal c and a flow variable (particularly the mixture fraction z)was hypothesized. To evaluate this, correlation coefficients were calculated using the methods described in [13] and [14]. However, the results indicate that there is no statistically meaningful correlation between the optimal c and the local mixture fraction z. This suggests that the effective contribution of $p_{\text{mean_skew}}$ does not directly depend on the mixture fraction value at a given cell.

Nonetheless, Figure 17 and Figure 30 reveal that cells in close proximity to the periodic boundary—regions dominated by nearly pure fuel—are better represented using $p_{\text{mean_var}}$ alone. Conversely, in the flame front and its adjacent zones, there is a noticeable tendency toward relying on $p_{\text{mean_skew}}$. Therefore, with further refinement and expansion of the dataset, it may become possible to establish a more robust correlation between the contribution of $p_{\text{mean_skew}}$ and specific flow variables such as mixture fraction or other combustion parameters.

Moreover, to evaluate the relative contributions of both $p_{\text{mean_var}}$ and $p_{\text{mean_skew}}$, the optimal c value is calculated and utilized. However, Equation 47 applies a linear interpolation between the two PDFs to estimate the combined behavior. While this approach is straightforward, it lacks the flexibility to accurately capture nuanced differences in the probability distribution. As illustrated in Figure Figure 21, even when the optimal c value is determined as $c = 0.7708$, the combined PDF fails to closely approximate the real PDF, particularly in the interval $z = 0.7$ to $z = 1.0$. This discrepancy arises despite the fact that the real PDF lies between $p_{\text{mean_var}}$ and $p_{\text{mean_skew}}$ in that region.

To address this limitation and further reduce the error defined in Equation 48, future work may im-

plement higher-order interpolation techniques such as polynomial interpolation ($p_{\text{comb}} = a \cdot p_{\text{meanvar}}^2 + b \cdot p_{\text{meanvar}} \cdot p_{\text{meanskew}} + c \cdot p_{\text{meanskew}}^2$). This would provide a smoother and more adaptable fit to the real PDF, potentially leading to enhanced accuracy in both error estimation and temperature prediction.

Finally, this study has demonstrated that incorporating higher-order moments of the mixture fraction—specifically through $p_{\text{mean_skew}}$—contributes meaningfully to the accuracy of the predictions. This finding is particularly valuable within the scope of the present work and establishes a foundational insight for future research directions.

Upon completion of the proposed future investigations on this topic, and provided that a robust correlation emerges between the contribution coefficient (denoted as c_{opt} in this work, and potentially generalized as a polynomial function in subsequent studies), these coefficients may be tabulated as a function of the mixture fraction. Such tabulations could then be applied to upcoming simulations to improve the prediction of both Favre-averaged temperature and species molar fractions.

References

[1] Michele Stefanizzi, Tommaso Capurso, Giovanni Filomeno, Marco Torresi, and Giuseppe Pascazio. Recent combustion strategies in gas turbines for propulsion and power generation toward a zero-emissions future: Fuels, burners, and combustion techniques. *Energies*, 14(20):6694, 2021. Published: 15 October 2021.

[2] Thierry Poinsot and Denis Veynante. *Theoretical and Numerical Combustion*. Edwards, Philadelphia, 3rd edition, 2012.

[3] Daniel Martinez-Sanchis, Andrej Sternin, Oskar Haidn, and Martin Tajmar. Turbulent combustion statistics in a diffusion flame for space propulsion applications. *Physics of Fluids*, 34(12):125115, 2022. Published online: 6 December 2022.

[4] Nikolaos Perakis, Christof Roth, and Oskar J Haidn. Development of a non-adiabatic flamelet model for reacting flows with heat loss. In *Space Propulsion Conference*, volume 2018, 2018.

[5] B. Ivancic, H. Riedmann, and M. Frey. Validation of turbulent combustion models for the 3d-simulations of liquid h2/o2 rocket combustors. In *Proceedings of the Space Propulsion Conference 2012*, Bordeaux, France, 2012.

[6] H. Riedmann, B. Kniesner, M. Frey, and C. Munz. Modeling of combustion and flow in a single element gh2/go2 combustor. *CEAS Space Journal*, 6:47–59, 2014.

[7] Norbert Peters. Laminar diffusion flamelet models in non-premixed turbulent combustion. *Progress in Energy and Combustion Science*, 10(3):319–339, 1984.

[8] Daniel Rahn, Hendrik Riedmann, Roland Behr, and Oskar J. Haidn. Non-adiabatic flamelet modeling for the numerical simulation of methane combustion in rocket thrust chambers. In *2018 Joint Propulsion Conference*, 2018.

[9] Stephen B. Pope. Pdf methods for turbulent reactive flows. *Progress in Energy and Combustion Science*, 11(2):119–192, 1985.

[10] D. C. Haworth. Progress in probability density function methods for turbulent reacting flows. *Progress in Energy and Combustion Science*, 36(2):168–259, 2011.

[11] Marvin Pommerening, Daniel Eiringhaus, Blazenko Ivancic, Chiara Manfletti, and et al. Advanced non-adiabatic flamelet modeling approach for transcritical combustion in liquid rocket engine thrust chambers. In *Space Propulsion Conference 2024*, Glasgow, Scotland, May 2024. Presented at Space Propulsion Conference 2024.

[12] Norman L. Johnson, Samuel Kotz, and N. Balakrishnan. Chapter 25: Beta distributions. In *Continuous Univariate Distributions, Volume 2*. Wiley, 2nd edition, 1995.

[13] Penn State University. 2.6 - (pearson) correlation coefficient r. https://online.stat.psu.edu/stat462/node/210/, 2021.

[14] Charles Spearman. The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1):72–101, 1904.

Appendix

Appendix A

ID	$Z_{\mathbf{real}}$	$Z_{\mathbf{prob}}$
1	NaN	0.00000
2	NaN	0.00000
3	NaN	0.00000
4	NaN	0.00000
5	0.09972	9.55e-05
6	0.11306	0.00198
7	0.13177	0.01331
8	0.15045	0.02112
9	0.16956	0.02194
10	0.18963	0.01703
11	0.20989	0.01282
12	0.22948	0.00988
13	0.24987	0.00820
14	0.27018	0.00641
15	0.28966	0.00463
16	0.30906	0.00348
17	0.32771	0.00264
18	0.34923	0.00332
19	0.36989	0.00297
20	0.39012	0.00225
21	0.40729	0.00237
22	0.43227	0.00283
23	0.44864	0.00308
24	0.46928	0.00264
25	0.49161	0.00294
26	0.51233	0.00270
27	0.53106	0.00331
28	0.55117	0.00405
29	0.56872	0.00547
30	0.58955	0.00458
31	0.60998	0.00524
32	0.62998	0.00601
33	0.65094	0.00578
34	0.66914	0.00369
35	0.68938	0.00436
36	0.71123	0.00593
37	0.73101	0.00700
38	0.75015	0.00685
39	0.77053	0.00863
40	0.78945	0.00739
41	0.81073	0.00804
42	0.83108	0.00669
43	0.84953	0.00958
44	0.87187	0.01183
45	0.88983	0.01352
46	0.91163	0.01568
47	0.92911	0.01452
48	0.94973	0.01465
49	0.97181	0.02312
50	0.99920	0.60559
51	1.00000	0.04986

Appendix A1. Discrete mixture fraction realizations and corresponding probabilities at cell (395, 1365).

ID	Z_{default}
1	0.00
2	0.02
3	0.04
4	0.06
5	0.08
6	0.10
7	0.12
8	0.14
9	0.16
10	0.18
11	0.20
12	0.22
13	0.24
14	0.26
15	0.28
16	0.30
17	0.32
18	0.34
19	0.36
20	0.38
21	0.40
22	0.42
23	0.44
24	0.46
25	0.48
26	0.50
27	0.52
28	0.54
29	0.56
30	0.58
31	0.60
32	0.62
33	0.64
34	0.66
35	0.68
36	0.70
37	0.72
38	0.74
39	0.76
40	0.78
41	0.80
42	0.82
43	0.84
44	0.86
45	0.88
46	0.90
47	0.92
48	0.94
49	0.96
50	0.98
51	1.00

Appendix A2. Ideal Z values matrix with uniform spacing of 0.02.

ID	$Z_{\mathbf{real}}$	$Z_{\mathbf{prob}}$
1	0.00000	0.00000
2	0.02000	0.00000
3	0.04000	0.00000
4	0.06000	0.00000
5	0.09972	9.55e-05
6	0.11306	0.00198
7	0.13177	0.01331
8	0.15045	0.02112
9	0.16956	0.02194
10	0.18963	0.01703
11	0.20989	0.01282
12	0.22948	0.00988
13	0.24987	0.00820
14	0.27018	0.00641
15	0.28966	0.00463
16	0.30906	0.00348
17	0.32771	0.00264
18	0.34923	0.00332
19	0.36989	0.00297
20	0.39012	0.00225
21	0.40729	0.00237
22	0.43227	0.00283
23	0.44864	0.00308
24	0.46928	0.00264
25	0.49161	0.00294
26	0.51233	0.00270
27	0.53106	0.00331
28	0.55117	0.00405
29	0.56872	0.00547
30	0.58955	0.00458
31	0.60998	0.00524
32	0.62998	0.00601
33	0.65094	0.00578
34	0.66914	0.00369
35	0.68938	0.00436
36	0.71123	0.00593
37	0.73101	0.00700
38	0.75015	0.00685
39	0.77053	0.00863
40	0.78945	0.00739
41	0.81073	0.00804
42	0.83108	0.00669
43	0.84953	0.00958
44	0.87187	0.01183
45	0.88983	0.01352
46	0.91163	0.01568
47	0.92911	0.01452
48	0.94973	0.01465
49	0.97181	0.02312
50	0.99920	0.60559
51	1.00000	0.04986

Appendix A3. Modified discrete mixture fraction realizations and corresponding probabilities at cell (395, 1365).